Module 4

Metagenomics application

INFORMATION

Lecturer

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Co-lecturer(s)

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Teaching assistant

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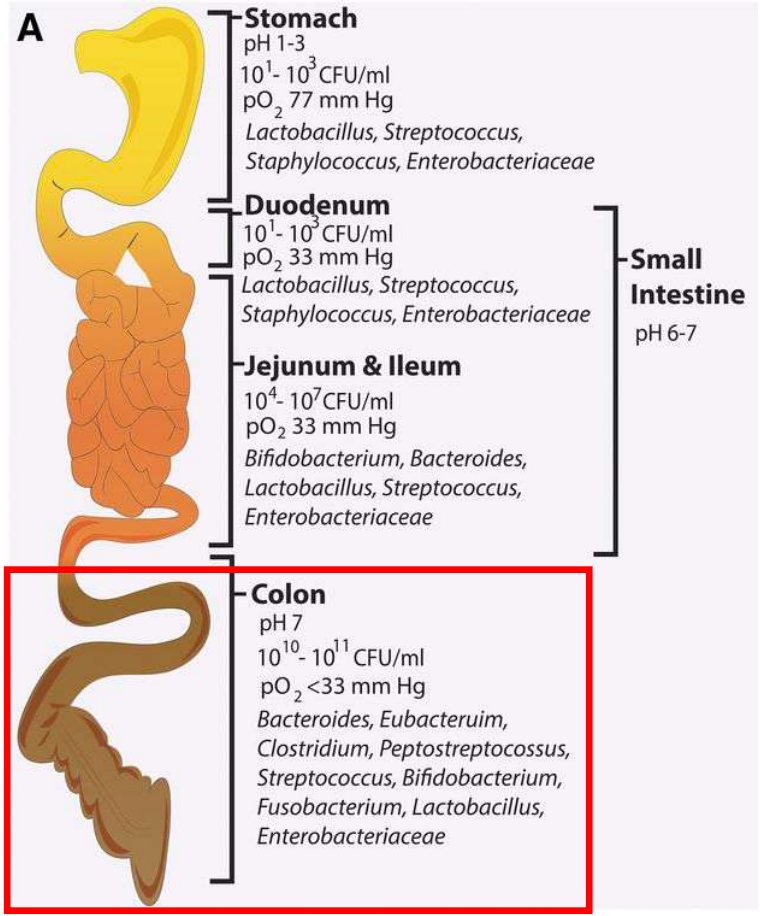
Agenda

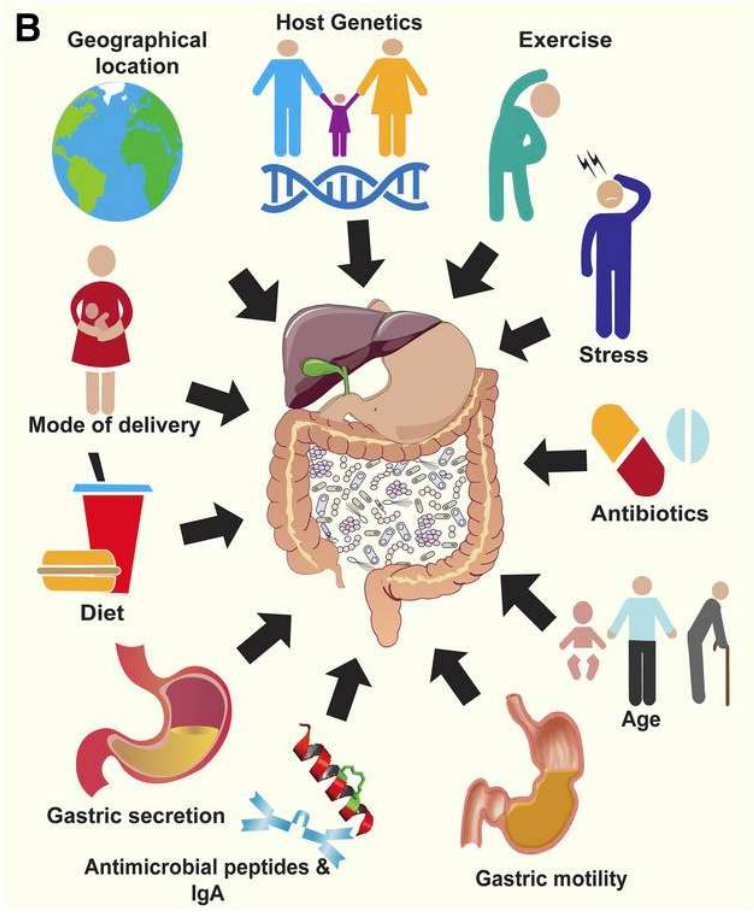
1. Microbiome and metagenomics
2. Applications
3. Notes on metagenomics
4. Metagenomic data analysis
5. Practical 1 – stool metagenomics
6. Practical 2 – metagenomics as diagnosis

The microbiome



- The colon microbiota harbors the greatest density of bacteria.
- The gut microbiota is affected by innumerable factors.

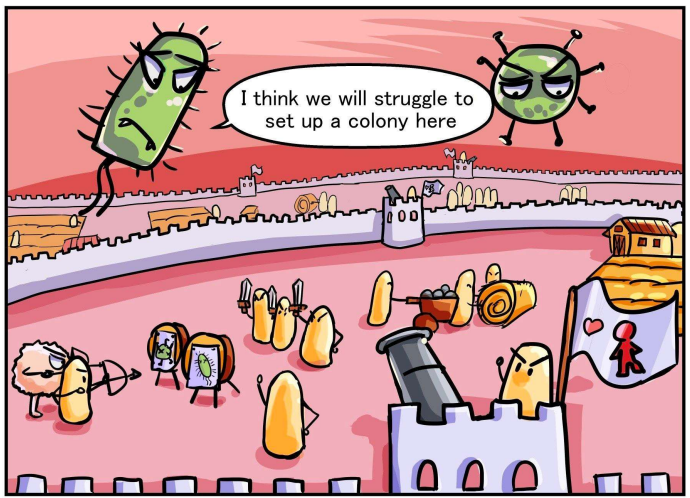


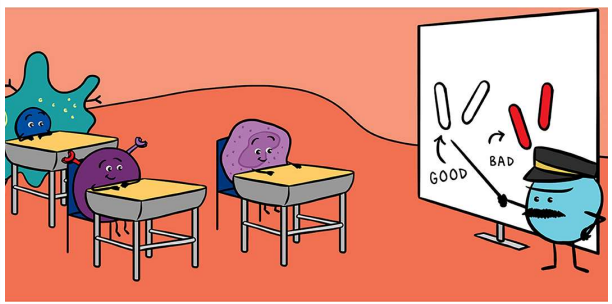


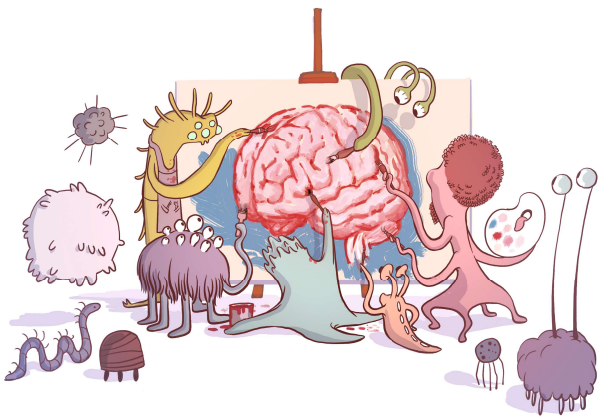
Clarke et al., **Pharmacol. Rev.**, 2019

Functions of the gut microbiome

- Food digestion (complex fibre)
- Colonization resistance
- Training of host immunity
- Producing bioactive compounds (short chain fatty acids)
- Regulating gut endocrine and neurological signaling (serotonin, GABA)







Methods of studying the microbiome

Metagenomics

16S rRNA sequencing

- Taxonomic profiling
- High throughput

Shotgun metagenomics

- Taxonomic and functional profiling
- Complex analytical tools

✓ Gene-focused
✓ Holistic





Culturing

Optimizing culturing conditions (media, anaerobic)

✓ Live bacteria
✓ Targeted

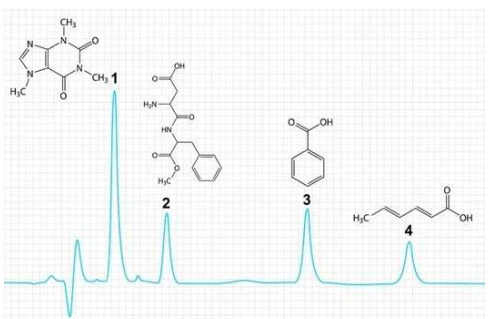


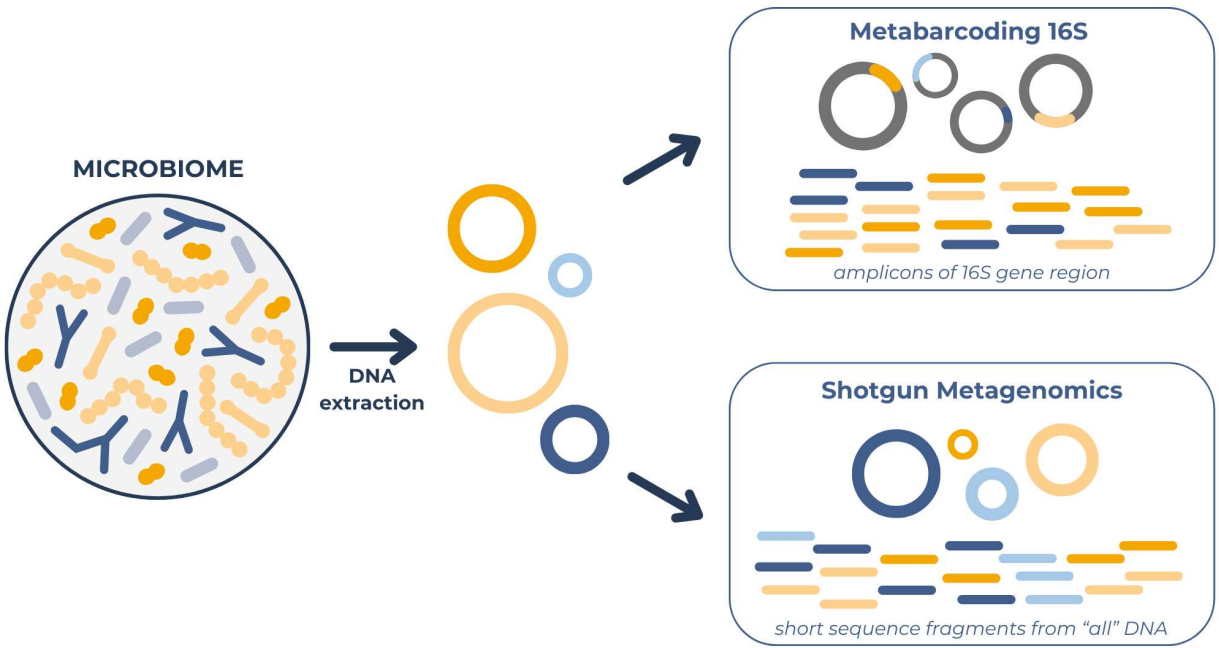
Metabolomics

Directly reflect the functional profile

Chemistry-based Chromatography – Mass spectrometry

✓ Metabolite focused
✓ Targeted





Computational analyses

Lower resource
Less complex
Require min 20,000 reads per sample

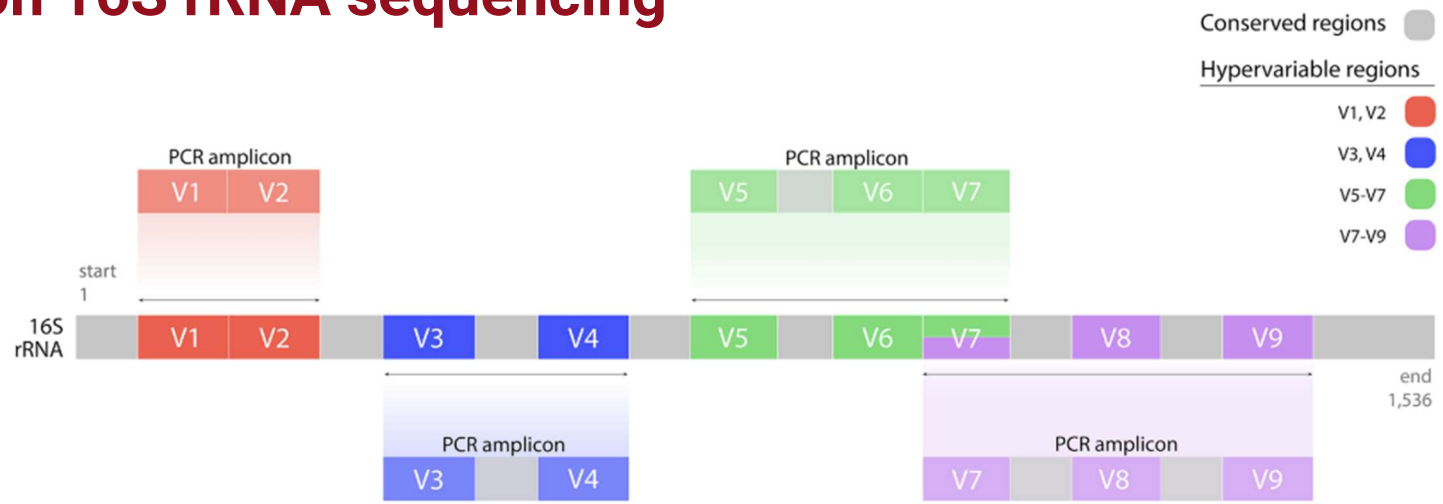
Higher resource
More complex
Requires min millions of reads

- Shallow: < 3M
- Moderate: 3M – 15M
- Deep: 15M – 60M
- Ultra-deep: > 60M

Feature	Targeted 16S rRNA/ ITS/ 18S rRNA	Shotgun (mNGS)
Turnaround time	20+ hours	20+ hours
Library preparation	Can be complicated	More streamlined
Organism coverage	Bacteria and/or fungi	Bacteria, fungi, parasites, viruses
Taxonomic resolution	Genus level	Species/strain level
Resistance gene detection	No	Yes
Host DNA interference	Minimal	Major challenge (75% - 95% output)
Sensitivity	Lower	Higher
Specificity	False positives possible	False positive possible
Cost (European reference)	Lower	Higher
Platform	Illumina , Nanopore	Illumina mostly

- Lack of standardised diagnostic criteria
- High analytical sensitivity → risk of detecting contamination
- **Require careful clinical interpretation and expert judgement**

A note on 16S rRNA sequencing



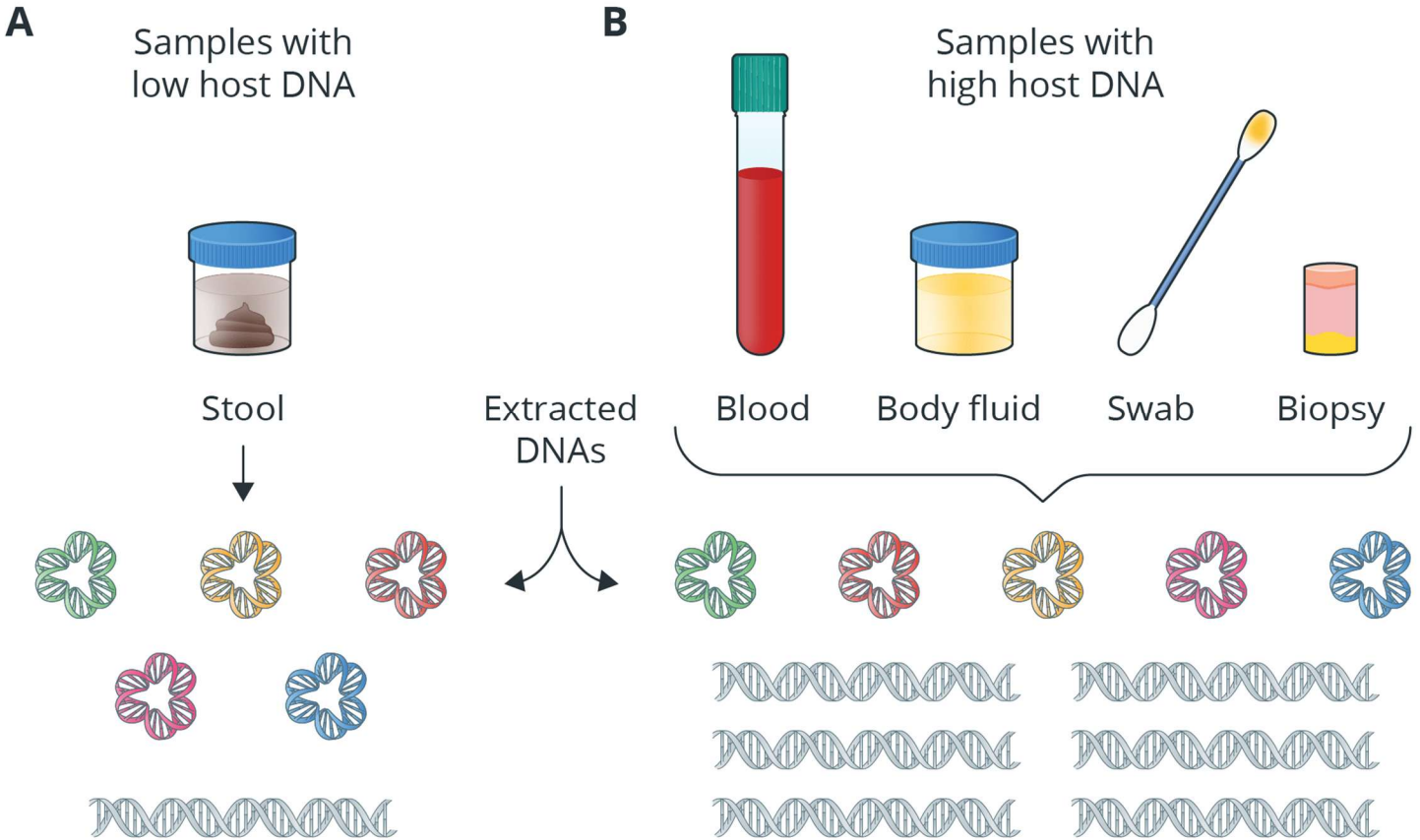
Oxford NANOPORE Technologies **DNA SEQUENCING TECHNOLOGIES** illumina

Oxford Nanopore: Direct Single-Molecule Sensing

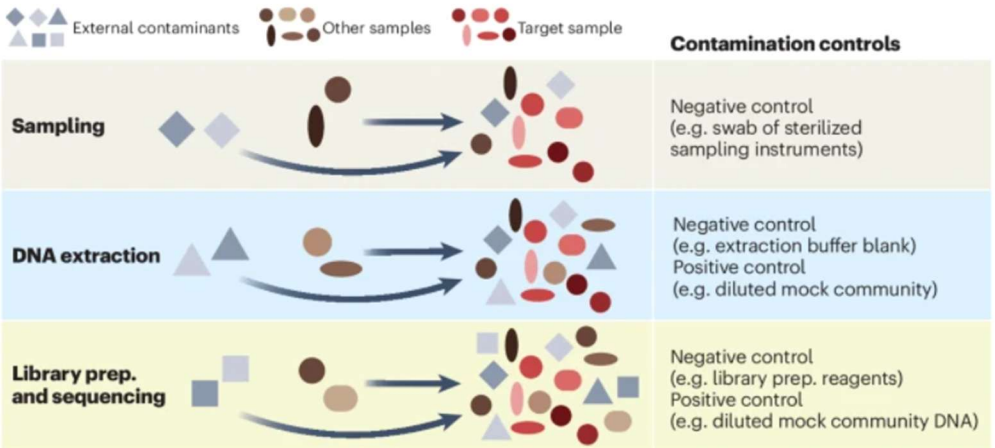
- Target full length 16S
- Low-throughput -> more costly
- Accuracy to subspecies level
- More rapid turnaround time, done locally

Illumina: Sequencing by Synthesis

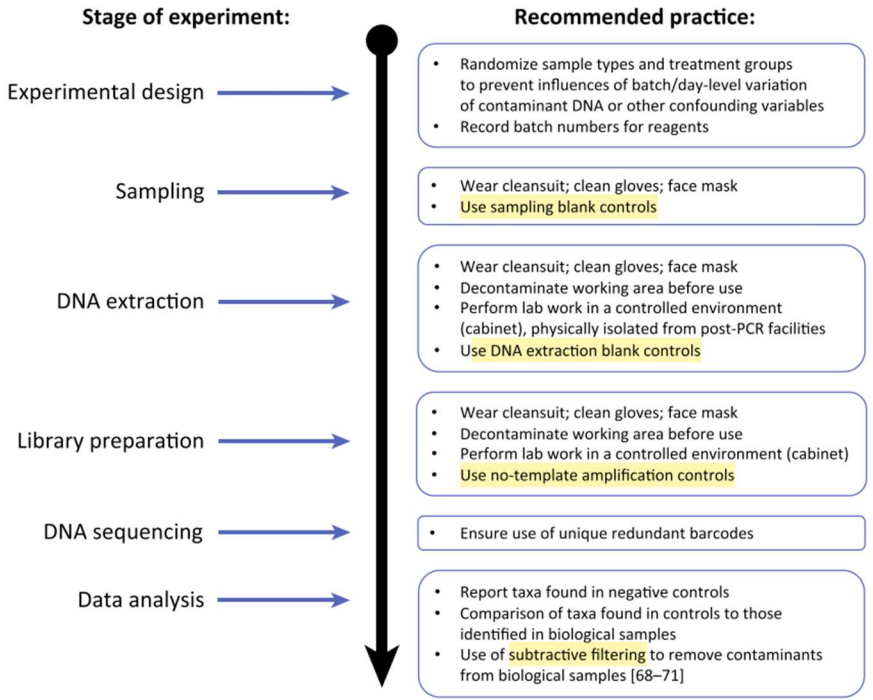
- Target individual V region (e.g. V3-V4)
- High-throughput -> lower cost per sample
- Accuracy mostly to the genus level
- Slow turnaround time, require sample pooling


A
Samples with
low host DNA
Stool
Extracted
DNAs
B
Samples with
high host DNA
Blood
Body fluid
Swab
Biopsy

Low biomass microbiome



Fierer et al., Nat Microbiology, 2025



Opinion

Contamination in Low Microbial Biomass Microbiome Studies: Issues and Recommendations

Eisenhofer et al., Trends in Microbiology, 2019

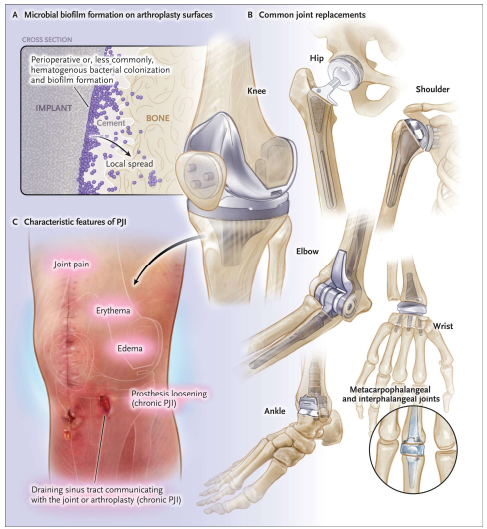
Common laboratory contamination genera *Ralstonia, Pseudomonas, Acinetobacter, Enhydrobacter*, etc.

Understand what you would expect from your samples

Clinical metagenomic

Clinical Infectious Diseases

MAJOR ARTICLE



Targeted Versus Shotgun Metagenomic Sequencing-based Detection of Microorganisms in Sonicate Fluid for Periprosthetic Joint Infection Diagnosis

Hyo-Lim Hong,[1,2] Laure Flurin,[1,3] Matthew J. Thoendel,[4] Matthew J. Wolf,[1] Matthew P. Abdel,[5] Kerryl E. Greenwood-Quaintance,[1] and Robin Patel[1,4]

Table 2. Comparison of Diagnostic Performance of Microbiologic Tests

Methods	Positive Number: PJI (n = 208)	Positive Number: NIAF (n = 187)	PPA (%) (95% CI)	NPA (%) (95% CI)
SF culture	110	0	52.9 (46.1–59.6)	100 (98.0–100)
Tissue culture	107	0	51.4 (44.7–58.2)	100 (98.0–100)
SF tNGS	150	2	72.1 (65.7–77.8)	98.9 (96.2–99.8)
SF sNGS	152	7	73.1 (66.7–78.7)	96.3 (92.5–98.2)

Abbreviations: CI, confidence interval; NIAF, noninfectious arthroplasty failure; NPA, negative percent agreement; PJI, periprosthetic joint infection; PPA, positive percent agreement; SF, sonicate fluid; sNGS, shotgun metagenomic sequencing; tNGS, 16S ribosomal RNA gene-based targeted metagenomic sequencing.

Table 4. Performance of Sonicate Fluid 16S Ribosomal RNA Gene-based Targeted Metagenomic Sequencing (tNGS) Versus Sonicate Fluid Shotgun Metagenomic Sequencing (sNGS)

Case Classification	Number of Samples	Identical Findings	Organisms Identified by tNGS but not sNGS	Organisms Identified by sNGS but not tNGS
NIAF	187	178 (95.2%)	2 (1.1%)	7 (3.7%)
Culture-positive PJI	110	92 (83.6%)	6 (5.5%)	13 (11.8%)[a]
Culture-negative PJI	98	76 (77.6%)	12 (12.2%)[b]	11 (11.2%)[c]

Data are the no. (%) of samples in which identical findings or discrepant findings between sonicate fluid tNGS and sonicate fluid sNGS were observed.

Abbreviations: NIAF, noninfectious arthroplasty failure; PJI, periprosthetic joint infection.

[a]Fourteen microorganisms detected by sNGS but not by tNGS in 13 cases.

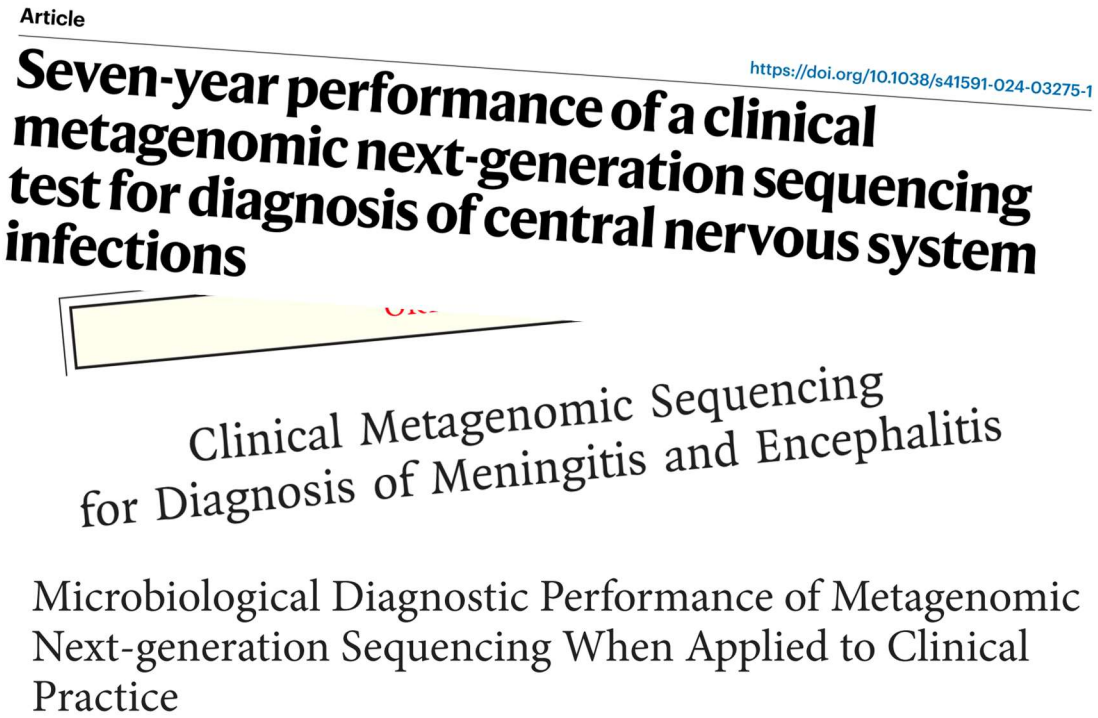
[b]Thirteen microorganisms detected by tNGS but not by sNGS in 12 culture-negative PJIs.

[c]Thirteen microorganisms detected by sNGS but not by tNorGS in 11 culture-negative PJIs.

- Both 16S and shotgun sequencing had similar performance in detecting pathogens in PJI
- Metagenomics detect organisms in 48% of culture-negative cases.

Hong et la., 2023, CID

Metagenomics (mNGS) has *increased sensitivity and high specificity* for diagnosis, but dependent on **specimens** and require both **bioinformatic and clinical interpretation**



Article

https://doi.org/10.1038/s41591-024-03275-1

Seven-year performance of a clinical metagenomic next-generation sequencing test for diagnosis of central nervous system infections

Clinical Metagenomic Sequencing for Diagnosis of Meningitis and Encephalitis

Microbiological Diagnostic Performance of Metagenomic Next-generation Sequencing When Applied to Clinical Practice

mNGS has increased diagnostic values for sterile fluids (CSF, pleural, joint, etc.)

What matters in clinical impact

Clinical Impact of Metagenomic Next-Generation Sequencing of Plasma Cell-Free DNA for the Diagnosis of Infectious Diseases: A Multicenter Retrospective Cohort Study

The mNGS Karius test results led to positive impact in 6 (7.3%), negative impact in 3 (3.7%), and **no impact in 71 (86.6%)**, and was indeterminate in 2 (2.4%) patients.

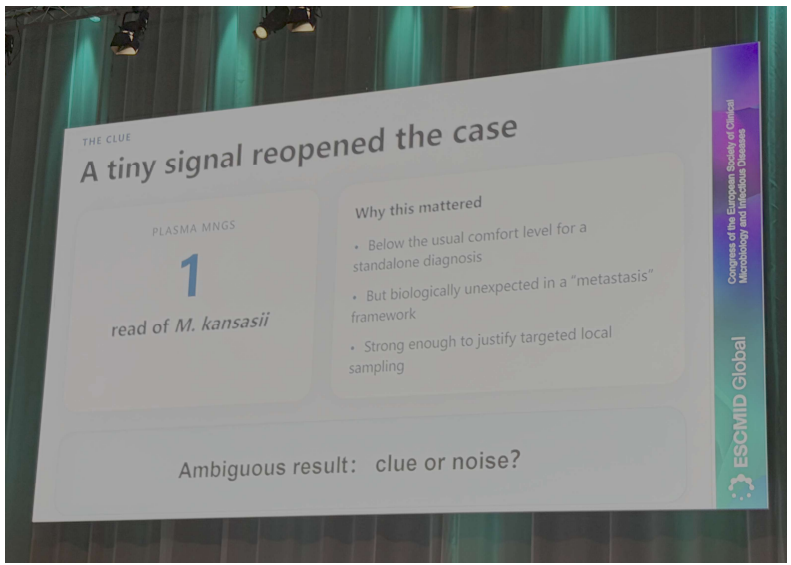
Recommendations for providers considering plasma mNGS testing for their patients.

Specific clinical syndromes where plasma mNGS testing has shown clinical utility
1. Culture-negative endocarditis or endovascular infections with blood cultures negative for 48-72 hours
2. Concern for zoonotic or vector-borne pathogens
3. Concern for certain fastidious pathogens (defined by ID)
Clinical scenarios where plasma mNGS testing may have clinical utility and may be considered, but more research studies needed to clarify the clinical impact
1. Pneumonia in immunocompromised patients (does not replace bronchoscopy or invasive tissue sampling given known sensitivity issue of plasma mNGS tests)
2. Invasive fungal infections
3. Complicated pneumonia in pediatric patients if no pathogen on respiratory specimen or pleural testing and if it will change management
4. Deep-seated abscess if aspiration from the source is not possible

UCLA internal recommendations for mNGS

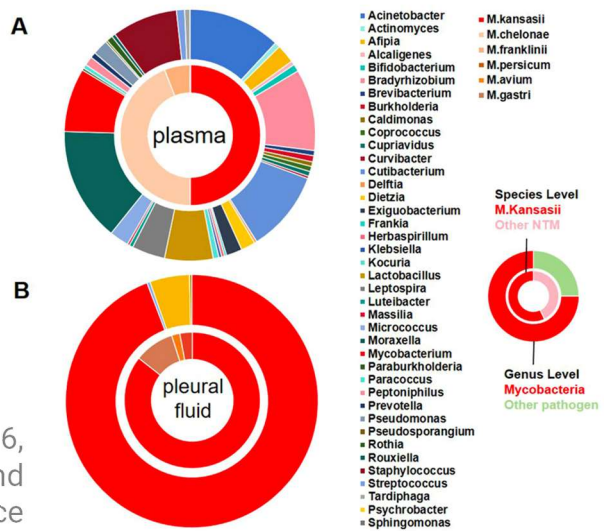
Metagenomics leading to clinical impact

Case Report: A 76-year-old woman with remote right breast carcinoma (mastectomy and adjuvant therapy in 1996) had a persistent right chest-wall lesion with rib changes and encapsulated pleural effusion repeatedly interpreted as metastatic disease from 2017 to 2023, despite multiple biopsies showing only fibrous hyperplasia. In August 2024, fever and cough prompted re-evaluation. PET-CT demonstrated a hypermetabolic pleura-adjacent lesion (SUVmax 10.8) without distant metastases. Plasma metagenomic next-generation sequencing (mNGS) yielded a low-level *M. kansasii* signal; pleural fluid mNGS identified 146 reads (94% relative abundance), later confirmed by culture. Targeted anti-NTM therapy stabilized the infection; however, the patient developed severe varicella-zoster virus infection and cardiac complications and subsequently died. The death was attributed to these complications rather than the progression of the *M. kansasii* infection.



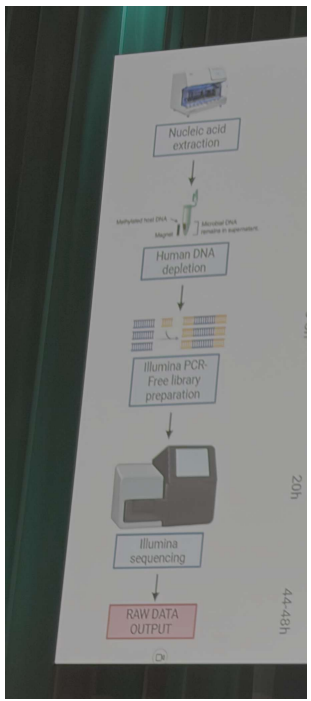
Credit: Dr. Qing Miao ESCMID 2026 presentation

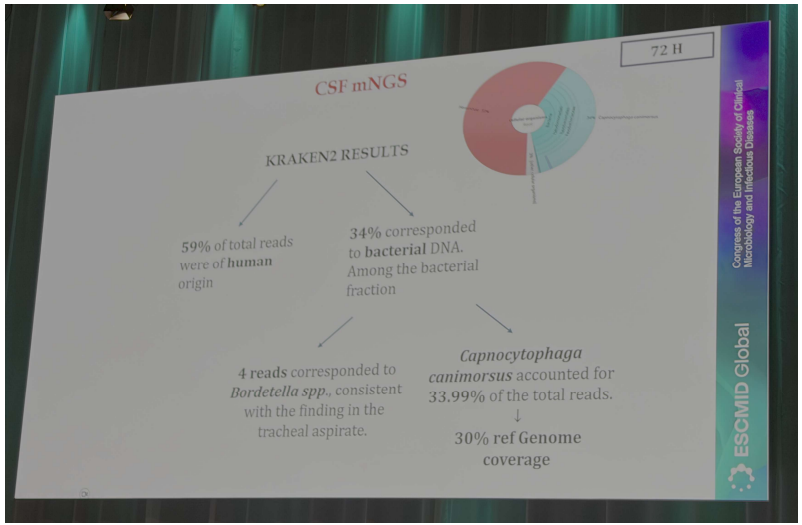
Plasma and pleural fluid mNGS results of the patient

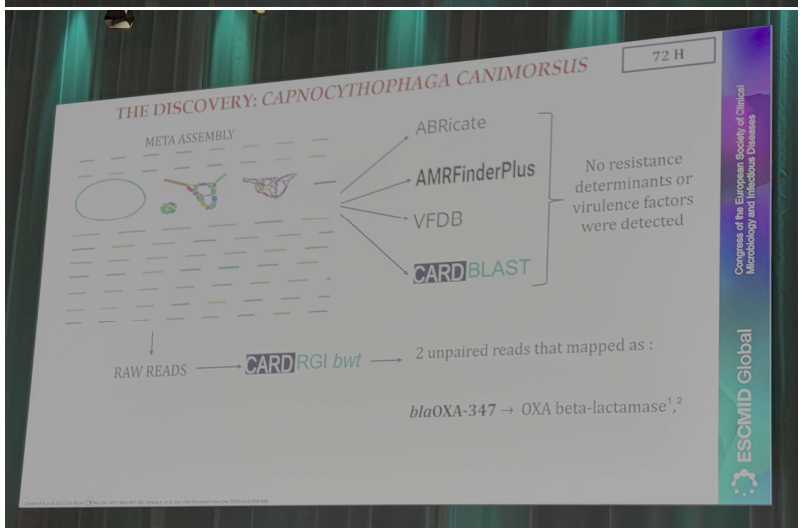


Fang et al., 2026, Infection and Drug Resistance

An Italian man diagnosed with meningitis







- Initiate *Capnocytophaga* treatment with beta-lactams
- Epidemiology linked to exposure to patient's own dog

- Noise or signal?
- Switch antibiotic therapy

Credit: Dr. Giovanni Lorenzin
ESCMID 2026 presentation

Brief summary

- Metagenomics is powerful for infectious disease diagnoses
- Both amplicon and shotgun metagenomics can be considered in real-world applications
- Clinical metagenomics is prone to contamination, leading to difficult interpretation
- Clinical metagenomics is specimen and clinical context dependent.
- Clinical metagenomics application needs **collaboration of clinical, microbiological and bioinformatics expertises**, to lead to impactful clinical decision.



Thank you!

Question?

Flower Thrower, Banksy

Beginner Course on Microbial Genomics: Applications & Research (2026)

Module 4 (cont')

Metagenomic data analysis

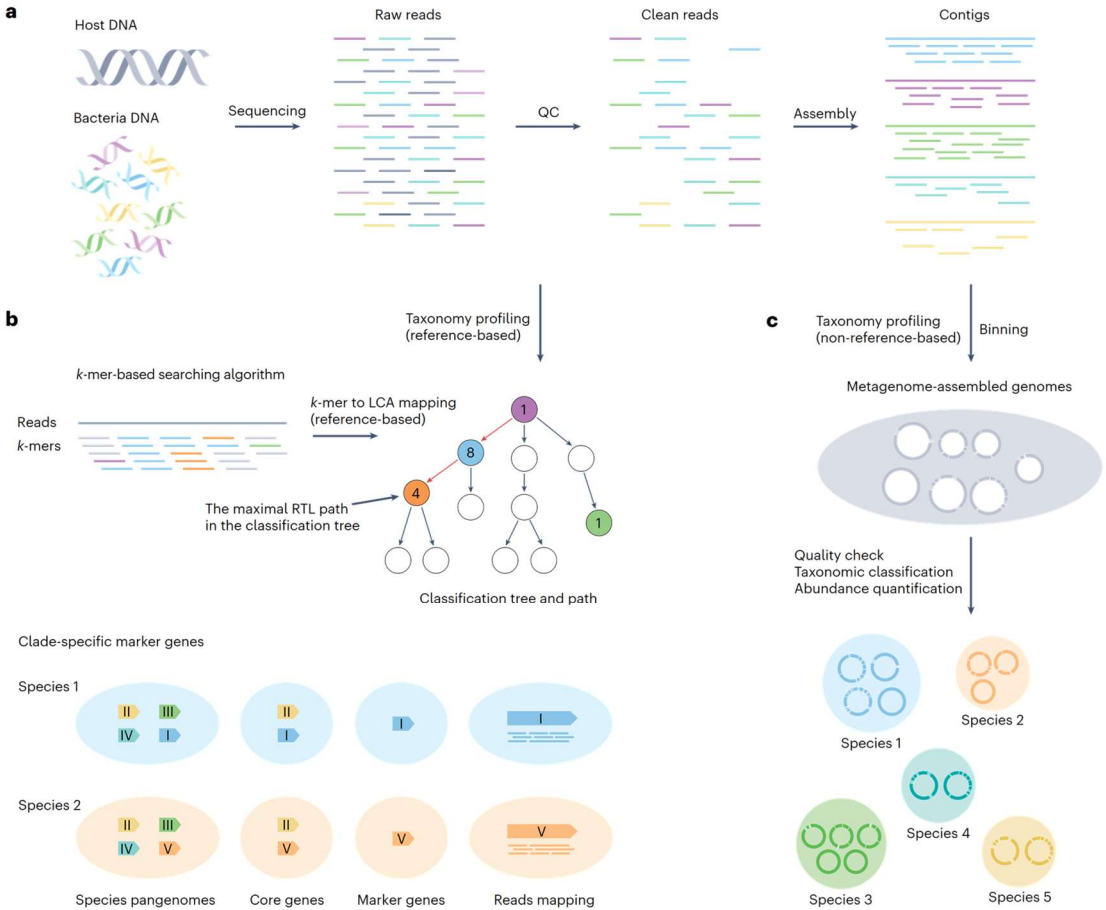
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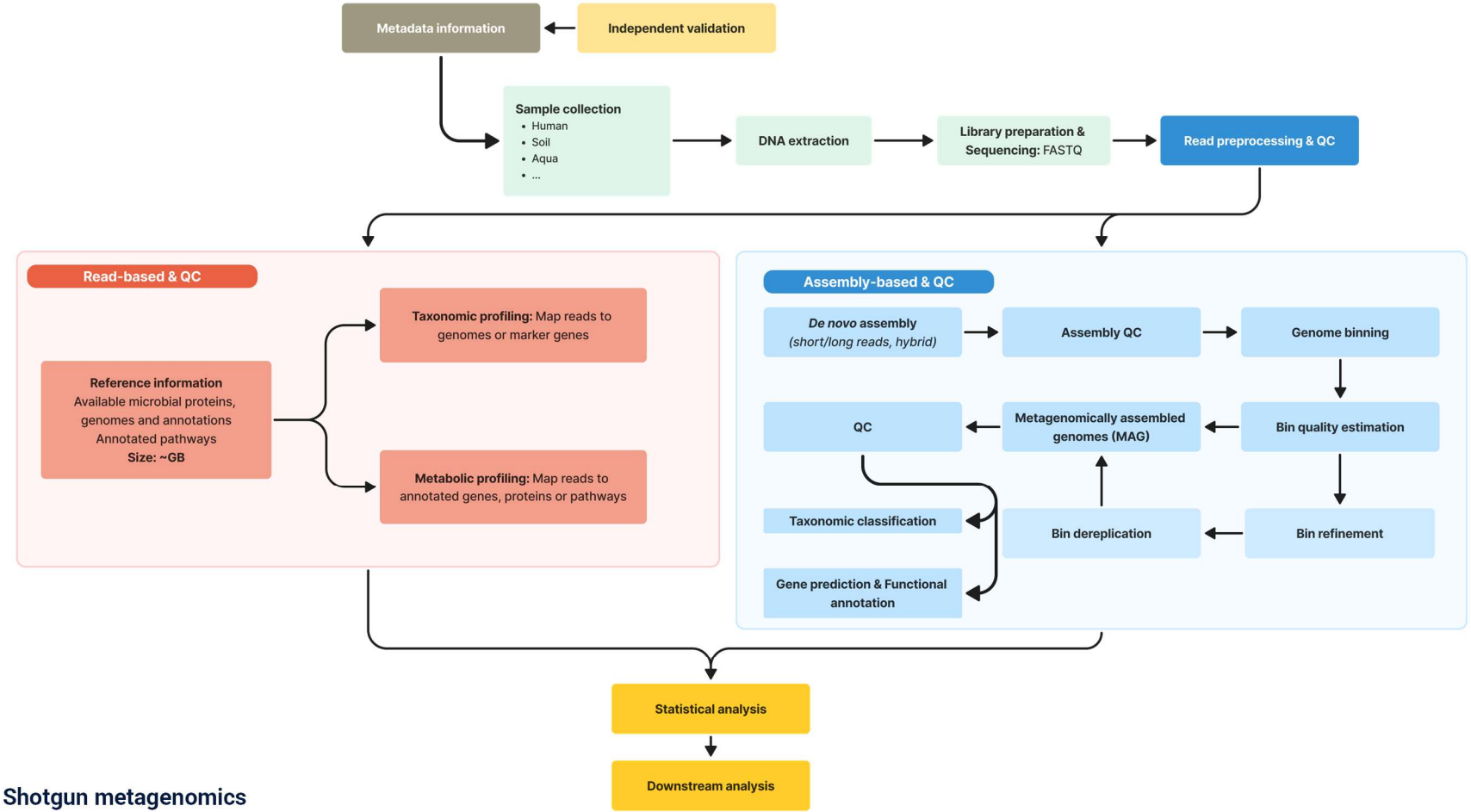
Teaching assistant
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Metagenomics workflow



Metagenomics processing and taxonomic profiling. a, Workflow of sequencing, quality control (QC; removing host contamination and low- quality reads) and reads assembly. **b,** Methods for reference-based taxonomic profiling. In the k-mer-based method (top), k-mers in the reads are mapped to the lowest common ancestor (LCA) in the database, and the maximal root-to- leaf (RTL) path score represents the classification path. In the clade-specific marker genes method (bottom), reads mapping to unique clade-specific marker genes recovered from over 1 million microbial genomes allow unambiguous taxonomic assignment. **c,** In non-reference taxonomic profiling by de novo assembly, metagenomic assemblies are grouped into bins to form metagenome-assembled genomes. Quality check, taxonomic classification and abundance quantification are then performed to retrieve detailed information for the metagenome-assembled genomes. *(10.1038/s44222-025-00346-x)*

Metagenomics workflow



OVERVIEW

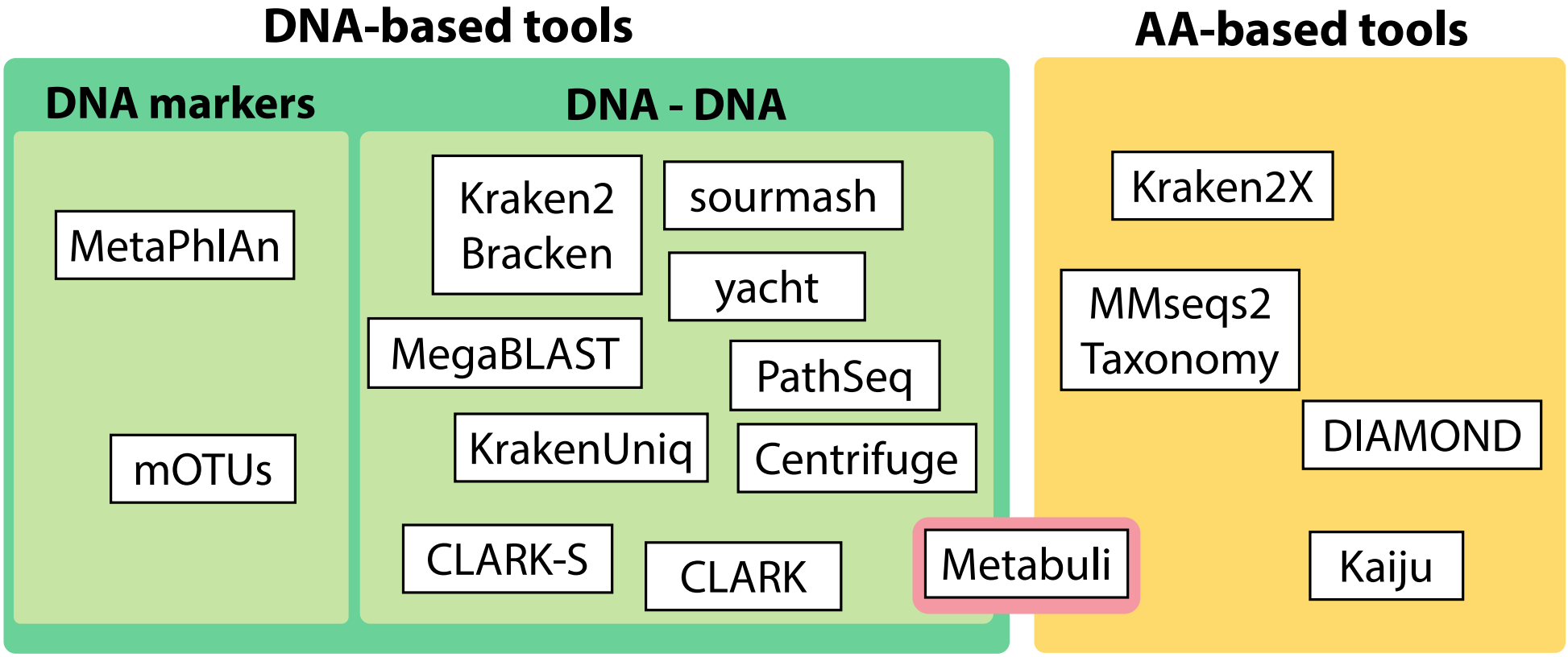
Read-based vs Assembly-based

Aspect	Read-based approach	Assembly-based approach
Core idea	Directly maps/classifies raw reads against reference databases	Assembles reads into contigs, then annotates genes or bins genomes
Main output	Taxonomic profiles, functional profiles, gene/pathway abundance	Contigs, genes, MAGs, genome-level functions
Speed & resources	Faster, lower computational demand	Slower, higher CPU/RAM/storage demand
Best for	Rapid screening, large cohorts, clinical or surveillance workflows	Genome reconstruction, novel taxa, strain/genome-resolved analysis
Strengths	Sensitive for known organisms and genes; works well for quick abundance estimation	Can recover new genomes and link genes to organisms
Limitations	Strongly depends on reference databases; limited novelty detection	May miss low-abundance organisms; assembly can be fragmented in complex samples

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What do we have now?



Trade-offs between sensitivity & specificity

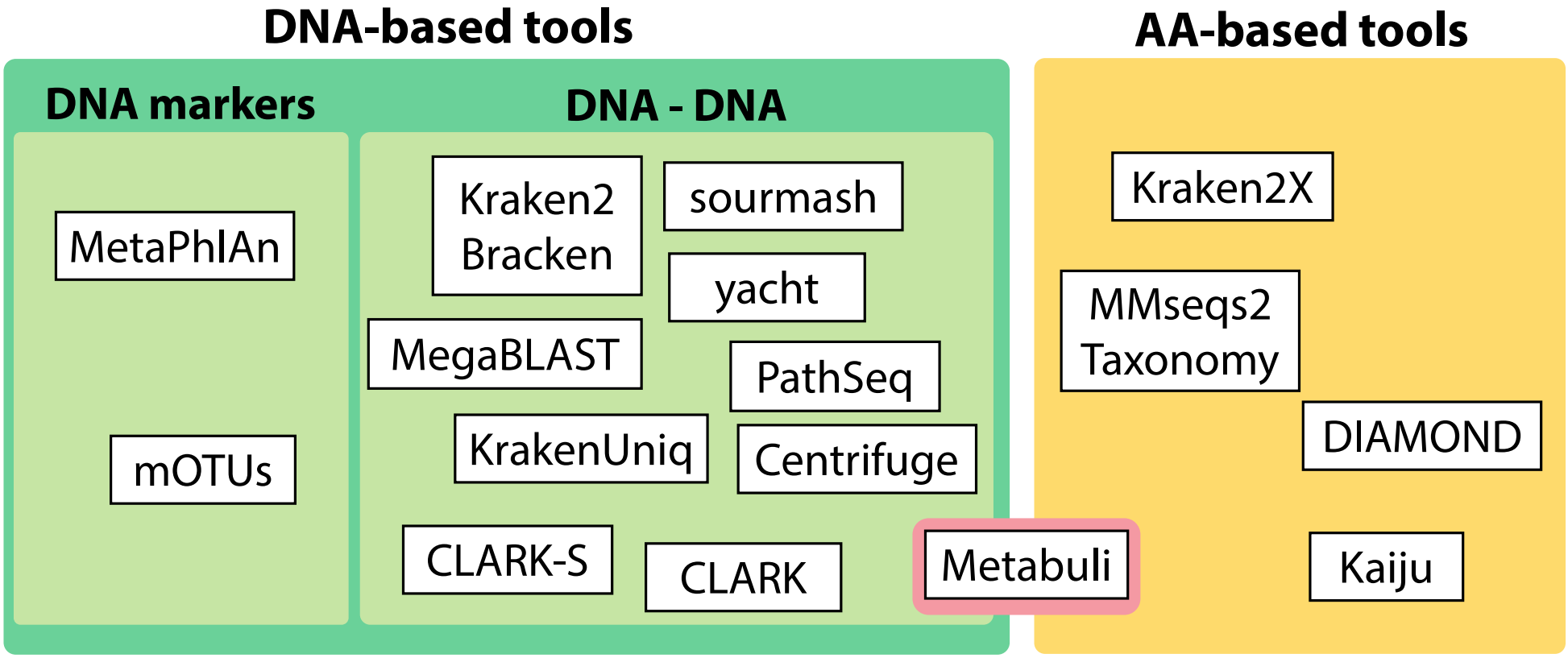
DNA vs. amino acid

Query		GCA	GCA	GCA	(Ala - Ala - Ala)
DB		GCT	GCA	GCA	(Ala - Ala - Ala)
		GCC	GCT	GCT	(Ala - Ala - Ala)

 is closer to query, but it cannot be resolved at the **amino acid level.**

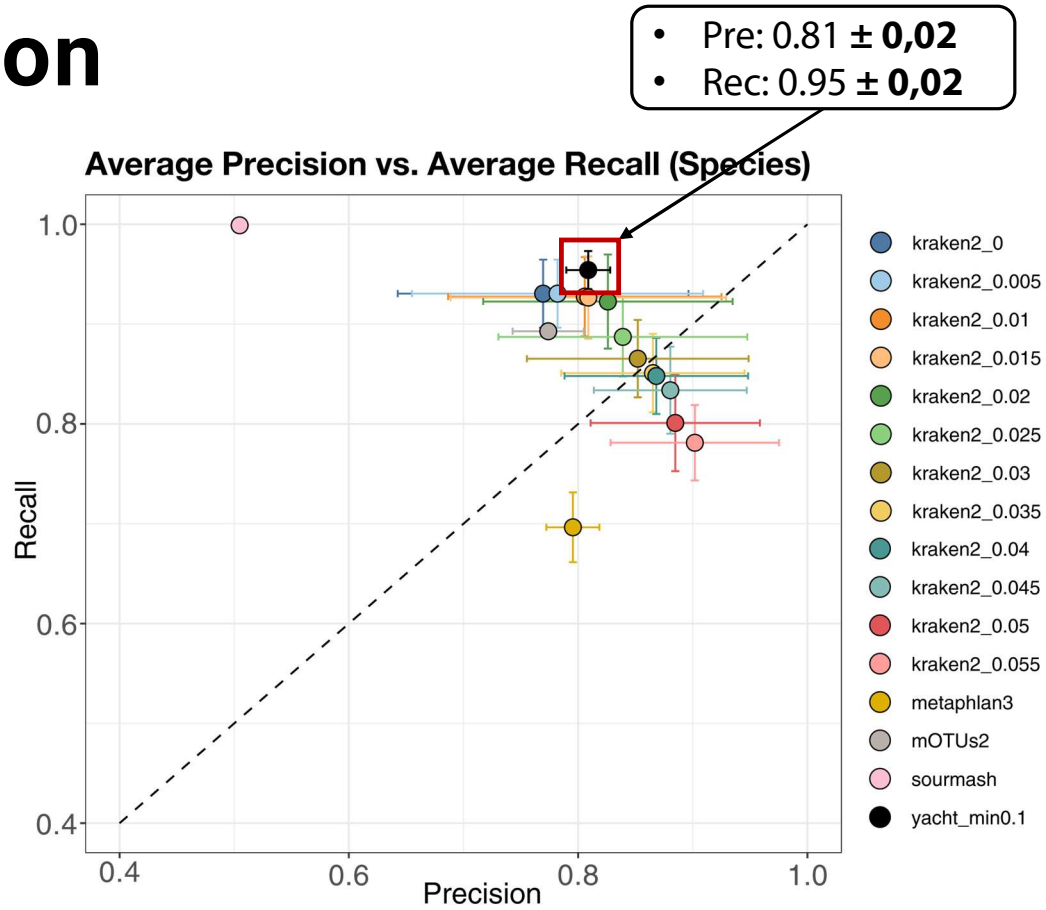
→ **lower specificity**

What do we have now?



Taxonomic classification

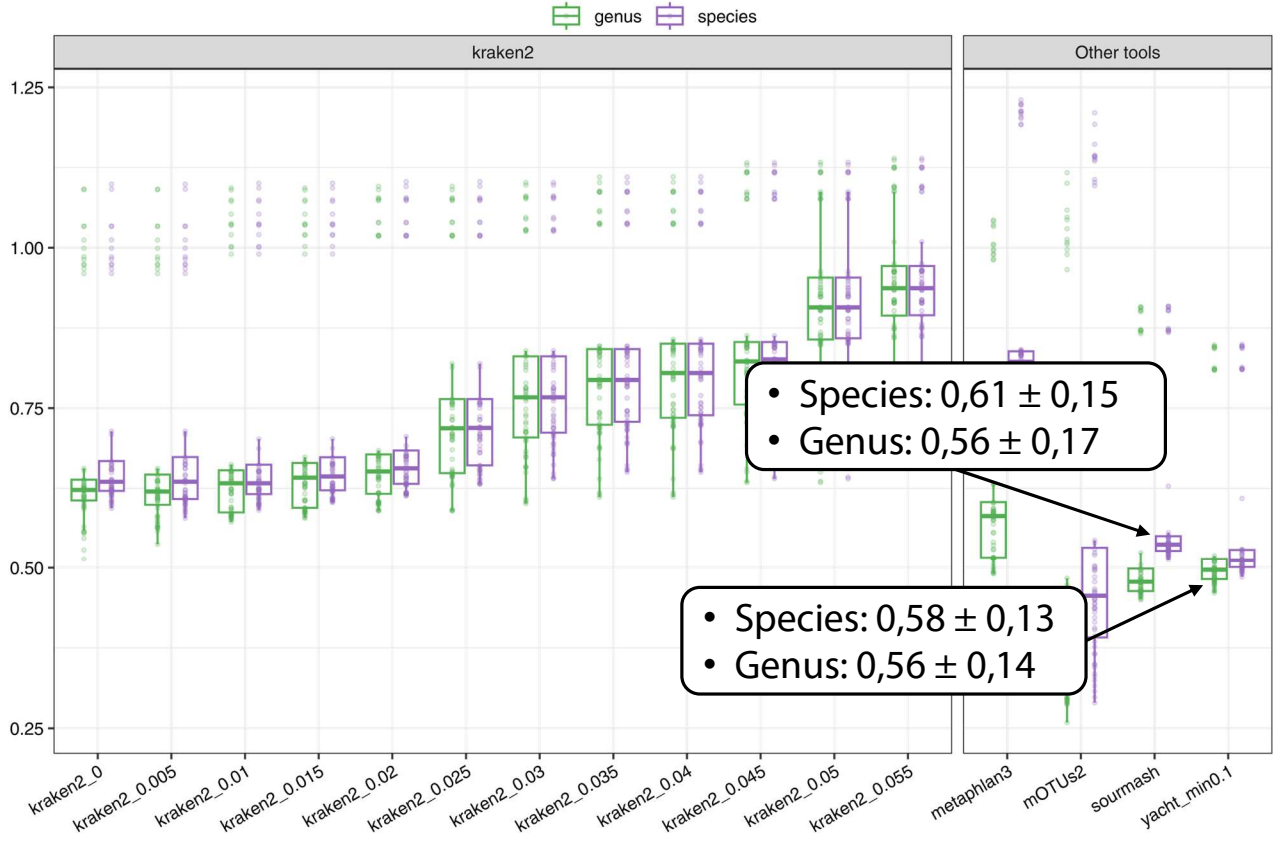
- Dataset: Tourlousse (2022) mock community (n = 56).
- Higher-right corner = better trade-off between precision and recall.



Precision–recall of 16 methods at species level; points and error bars show mean ± SD *(unpublished data)*

Taxonomic reconstruction accuracy

- Higher values indicate greater deviation from the original reference dataset.



Ability of each method to accurately reconstruct the microbial composition within a sample. Results are shown for 16 different methods evaluated on the Tourlousse dataset (n = 56) at the genus and species taxonomic levels. *(unpublised data)*

Resource usage (standard conditions)

Công cụ	Tiến hành lần lượt tổng cộng 140 mẫu		
	Bộ nhớ tối đa (GBs)	CPU (%)	Tổng thời gian (h)
Kraken 2_0	94,456	251	5,35
Kraken 2_0.005	94,419	251	5,24
Kraken 2_0.01	94,437	258	5,34
Kraken 2_0.015	94,434	244	5,37
Kraken 2_0.02	94,418	258	5,27
Kraken 2_0.025	94,406	253	5,63
Kraken 2_0.03	94,403	262	5,23
Kraken 2_0.035	94,443	254	6,37
Kraken 2_0.04	94,405	250	5,22
Kraken 2_0.045	94,387	254	6,17
Kraken 2_0.05	94,419	251	5,24
Kraken 2_0.055	94,428	220	5,99
MetaPhlAn 3	4,923	1322	11,62
sourmash	2,859	383	15,17
yacht	15,237	265	71,54
mOTUs2	4,857	155	8,7

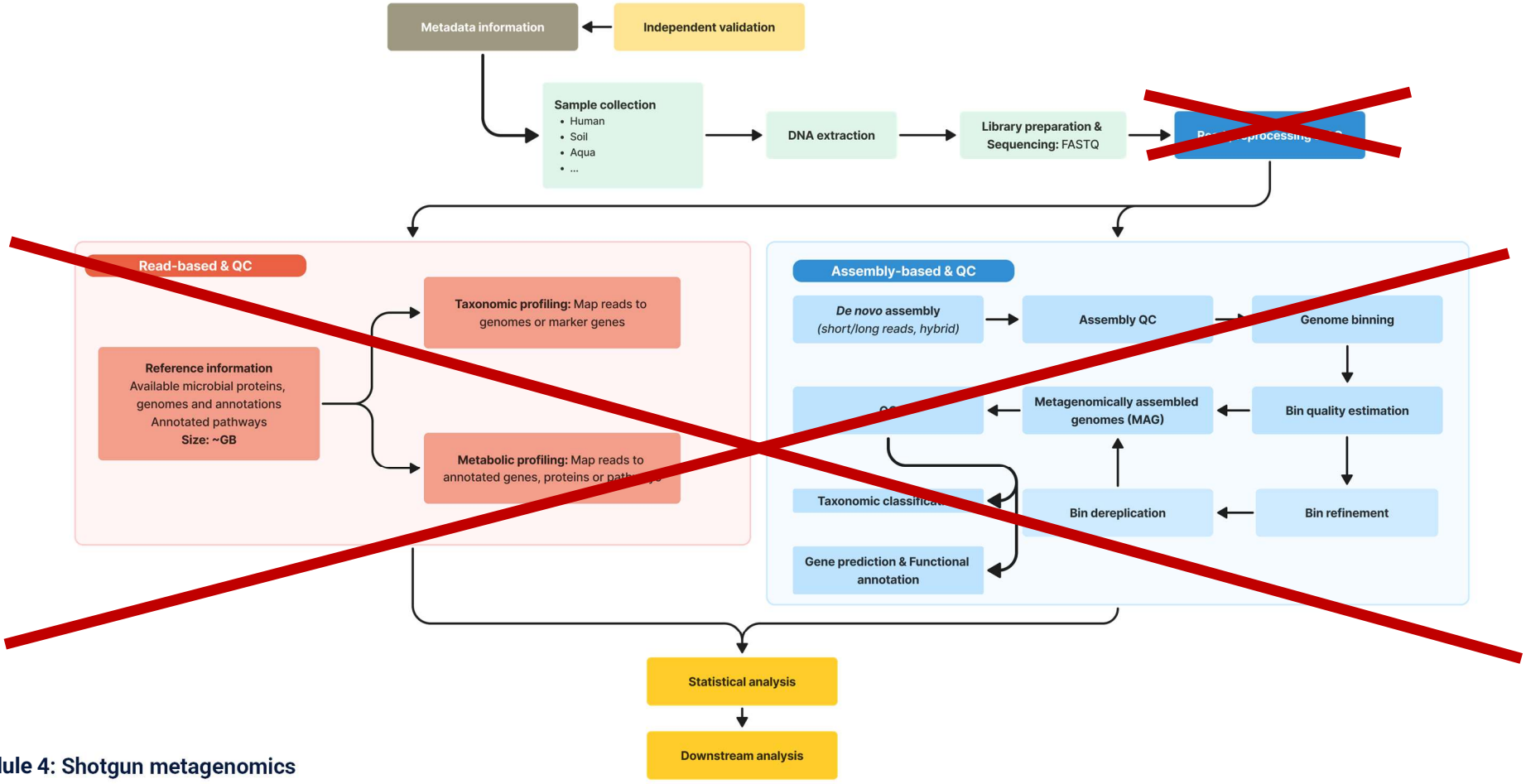
Performance and computational resource consumption of the evaluated tools under standard operating conditions, including runtime and memory usage measurements.

How much depth is enough?

Study goal	**Reasonable depth based on** Treichel, N. S., et al. (2026). *Nature Microbiology*	**Interpretation**
Known taxa / known strain detection with good references	**0.5-1 Gb**	Suitable for large cohorts and surveillance, but database quality is critical.
Relative abundance profiling	**0.5-2 Gb**	Often stable, but low-abundance taxa and wet-lab bias remain concerns.
KEGG pathway-level functional profiling	**2-5 Gb**	Reasonable for broad metabolic capacity, not for detailed gene recovery.
Protein repertoire / accessory gene recovery	**≥10 Gb**, often more	Strongly dependent on community complexity and abundance distribution.
De novo MAG reconstruction	**>10 Gb**, preferably with long reads/hybrid support	Standard HQ metrics are insufficient; chimerism must be explicitly tested.
Host-rich or low-biomass samples	**Depth must be adjusted upward**	Raw depth is not enough; microbial fraction and library prep dominate reliability.

DIVING INTO THE PIPELINE - metaflow

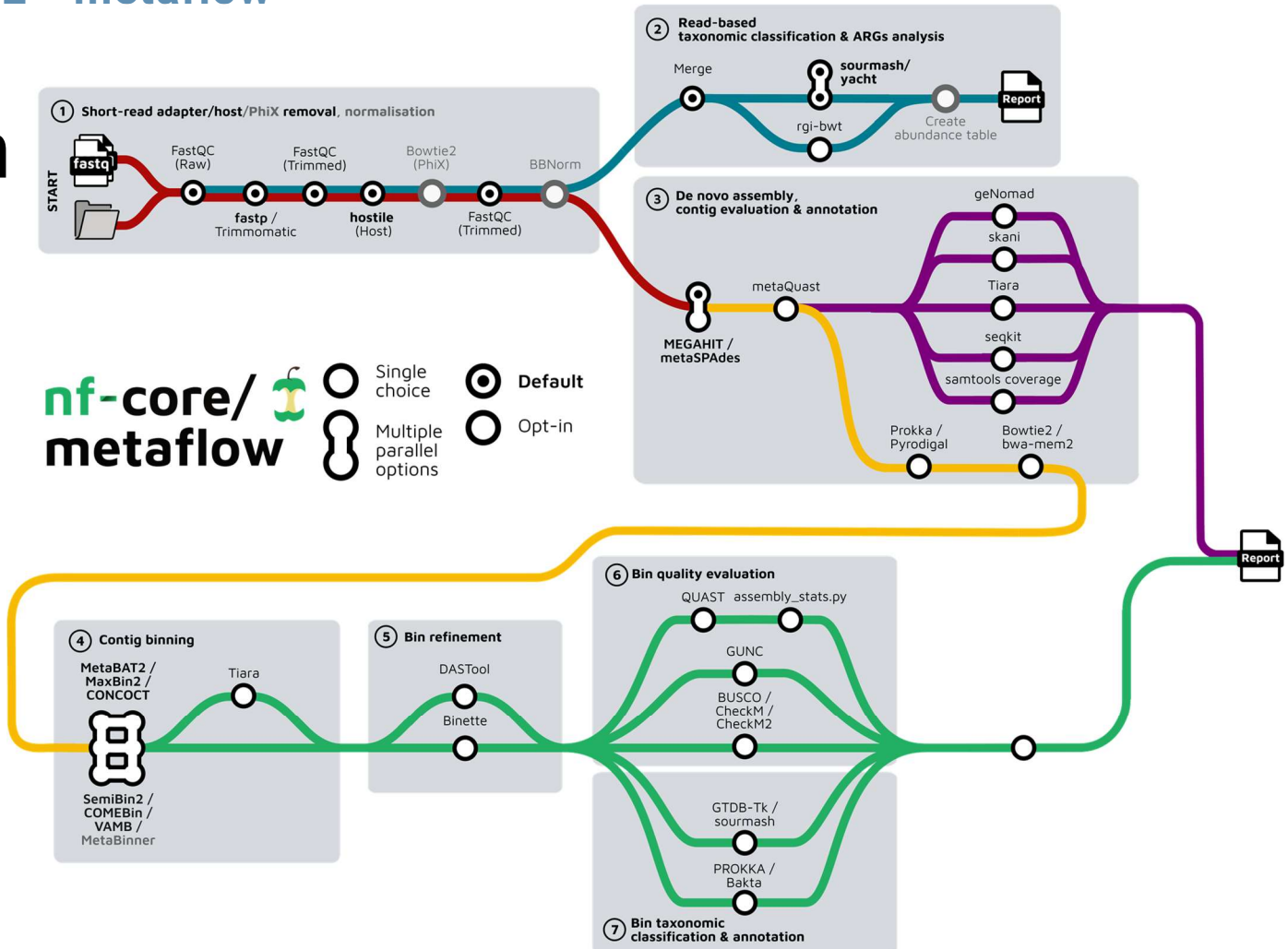
“There must be an easier way!”



Goals

- Core goal: Design and optimize a robust, scalable, and reproducible pipeline for comprehensive metagenomic analysis.
- Two core strategies for deep insights
  - **Assembly-based:** Genome binning for novel genome discovery and detailed analysis.
  - **Read-based:** Rapid analysis using raw reads.

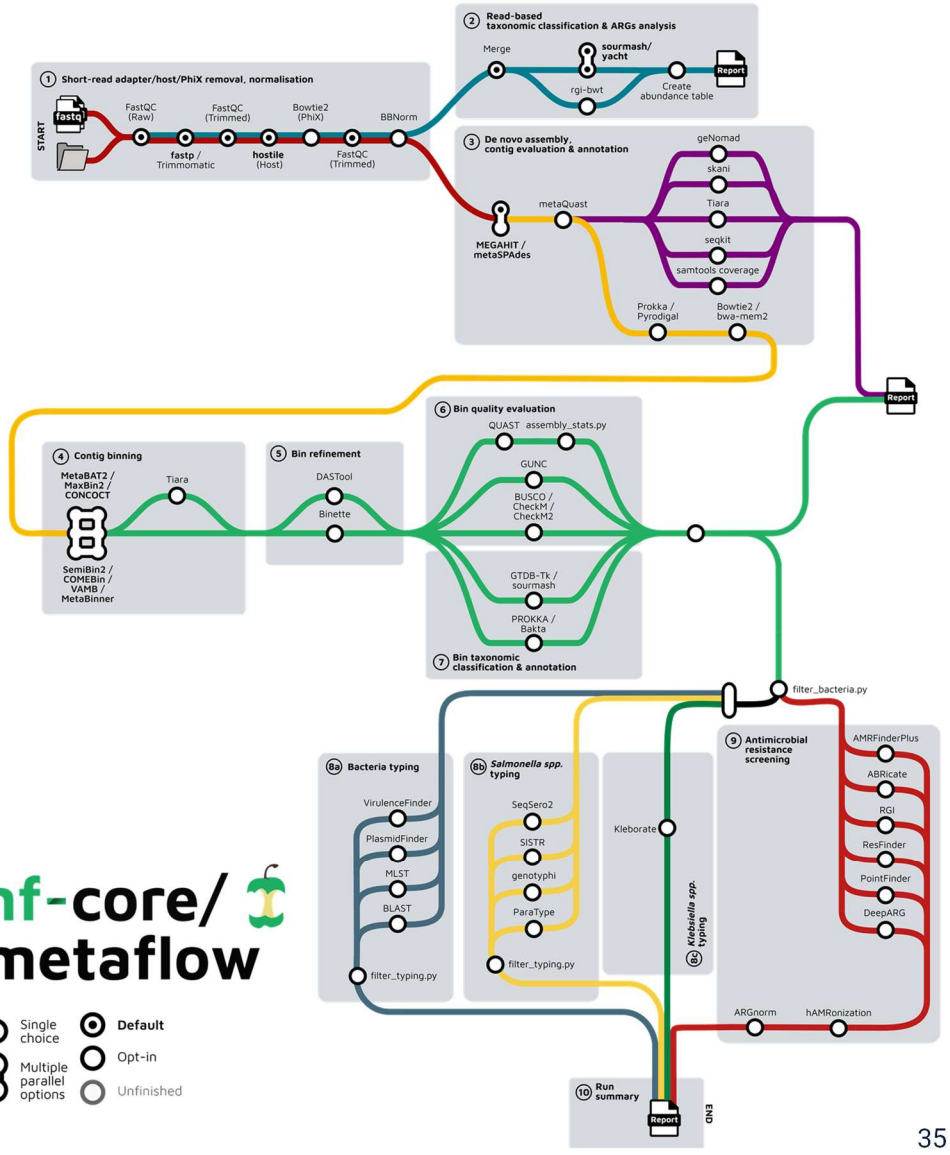
metaflow – current version



Future-proofing

- Support long-read sequencing.
- Support abundance table for differential abundance analysis.
- Typing: bacteria, *Salmonella*, *Klebsiella*, *Enterobacter*.
- AMR screening.
- Plasmid recovery?

metaflow: Roadmap



nf-core/ metaflow

Single choice
Multiple parallel options
Default
Opt-in
Unfinished

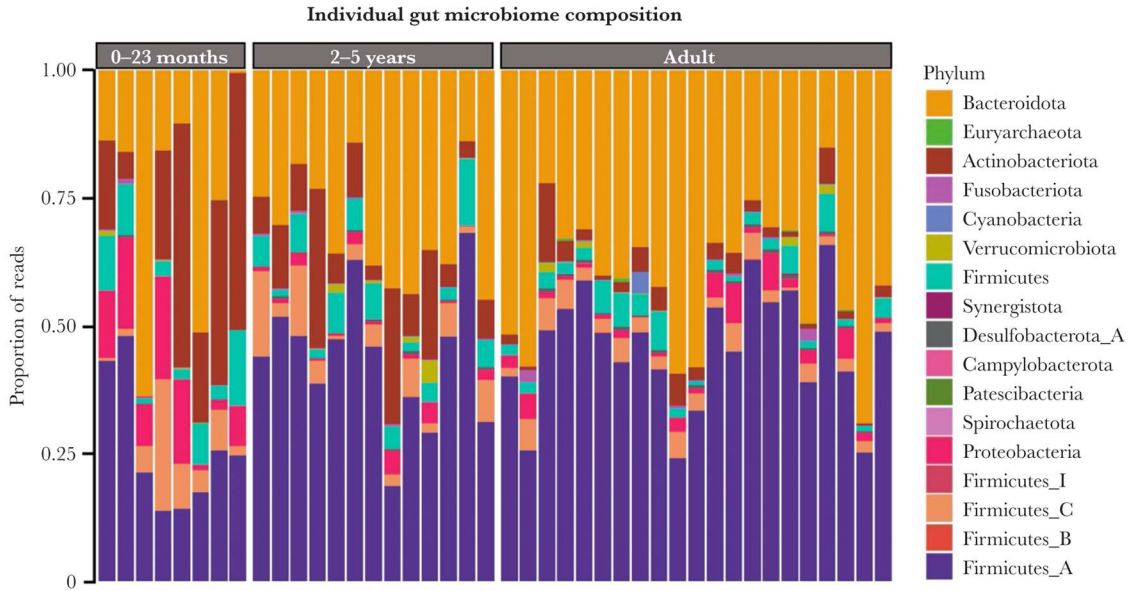
Wiki & Documentation

- Setup & DB prep: https://github.com/tnmquann/metaflow/wiki
- Nextflow manual: https://www.nextflow.io/docs/latest/index.html

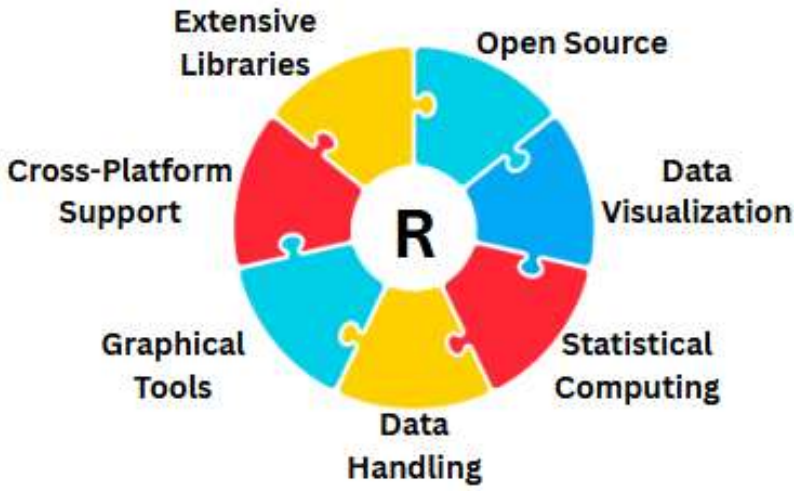
Practical 1 – stool microbiome

The Gut Microbiome of Healthy Vietnamese Adults and Children Is a Major Reservoir for Resistance Genes Against Critical Antimicrobials

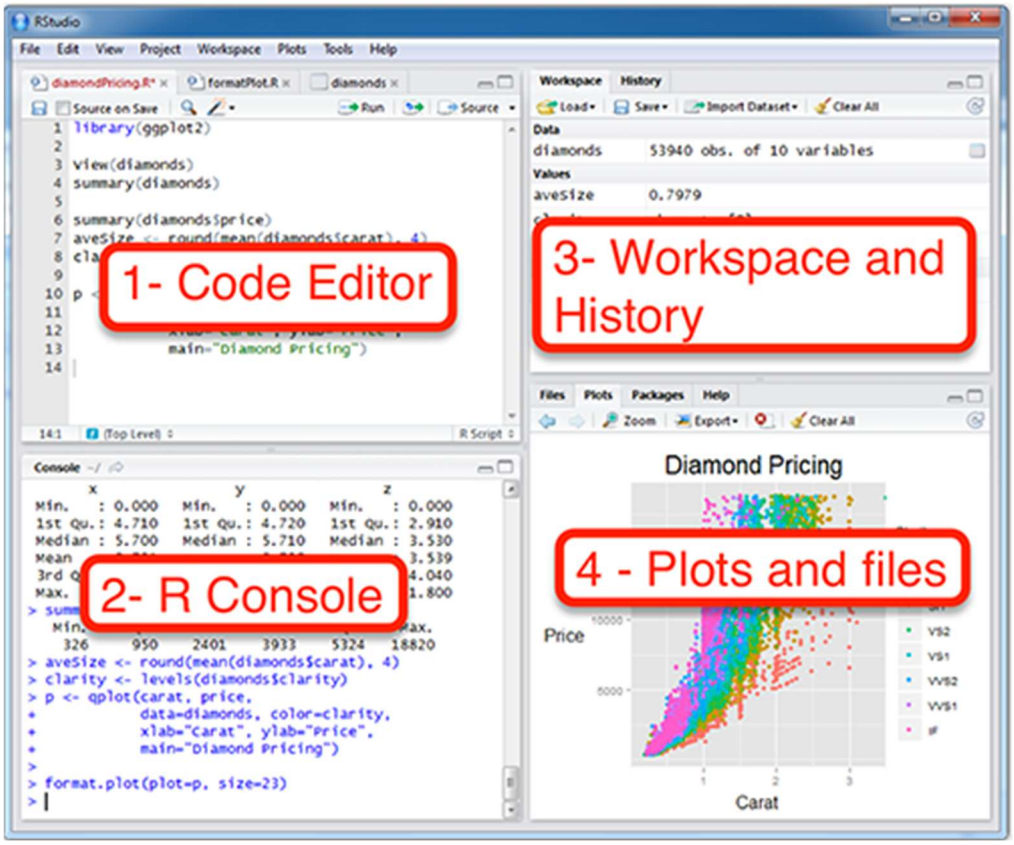
Joana Pereira-Dias,[1,2] Chau Nguyen Ngoc Minh,[3] Chau Tran Thi Hong,[3] To Nguyen Thi Nguyen,[3] Tuyen Ha Thanh,[3] Caroline Zellmer,[1,2] Hao Chung The,[3] Lindsay Pike,[4] Ellen E. Higginson,[1,2] and Stephen Baker[1,2]



Introducing R and RStudio



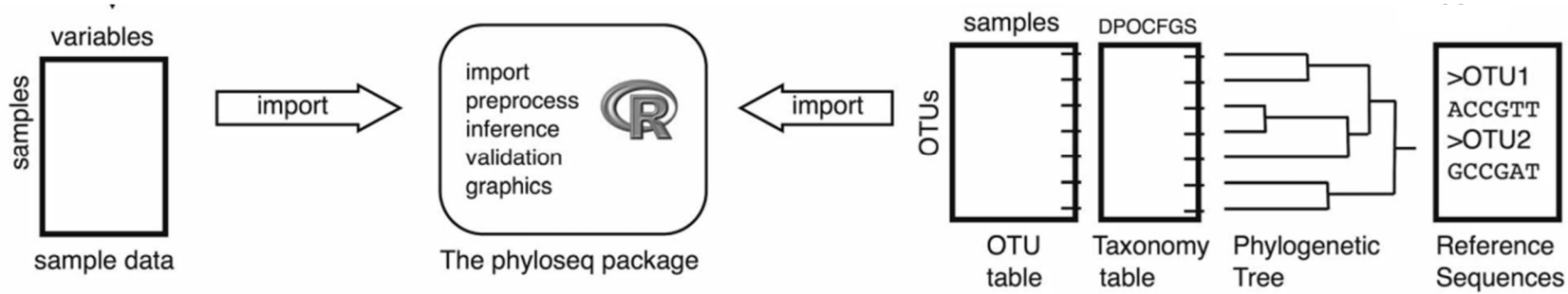
- R is like Excel on steroid!
- Mainly useful for complex data handling, pretty graphs, and robust statistics.
- A large community of developers for diverse fields in medical sciences.



Phyloseq integrates microbiome related data

What should be input:

- OTU table: directly from ASV table after removal of nonbacterial taxa
- Taxonomy table: output from ‘assignTaxonomy’ function
- Phylogenetic tree: Read in tree built separately
- Sample metadata: csv file with rownames() matching sample_names()



phyloseq: An R Package for Reproducible Interactive Analysis and Graphics of Microbiome Census Data

Paul J. McMurdie, Susan Holmes*

McMurdie and Holmes, 2013, PLoS One → 19,900 citations

Diversity indices

Alpha-diversity: for within-sample calculation

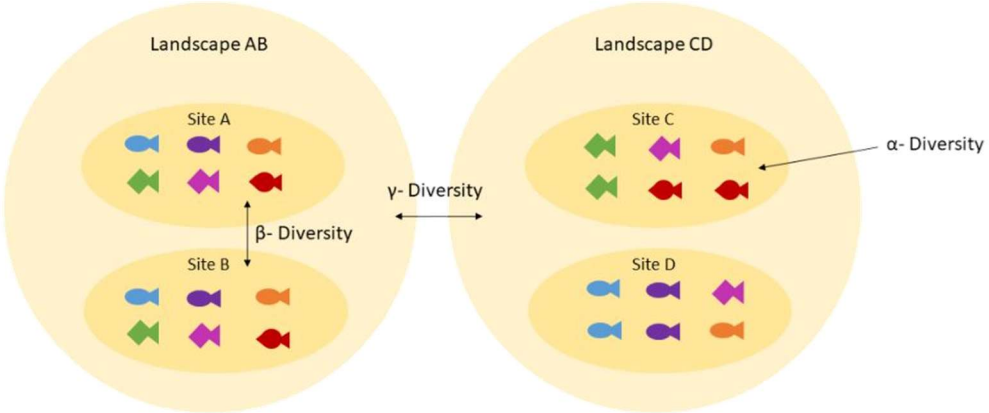
- Richness: number of taxa present in the sample
- Shannon index: combining richness with evenness

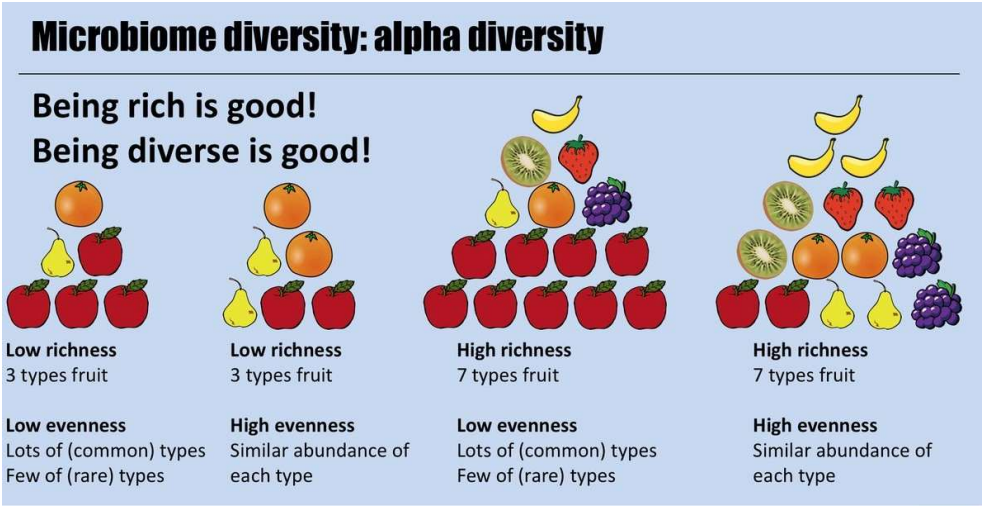
Beta-diversity: between-sample calculation

- Phylogenetic unaware: Bray-Curtis
- Phylogenetic aware: Unifrac, Weighted Unifrac, phylogenetic ILR (PhILR)

$$H' = -\sum_{i=1}^{S} p_i \cdot \ln p_i$$

Credit: VMI Bootcamp
https://awbrooks19.github.io/vmi_microbiome_bootcamp/rst/4_concepts_of_community_analysis.html

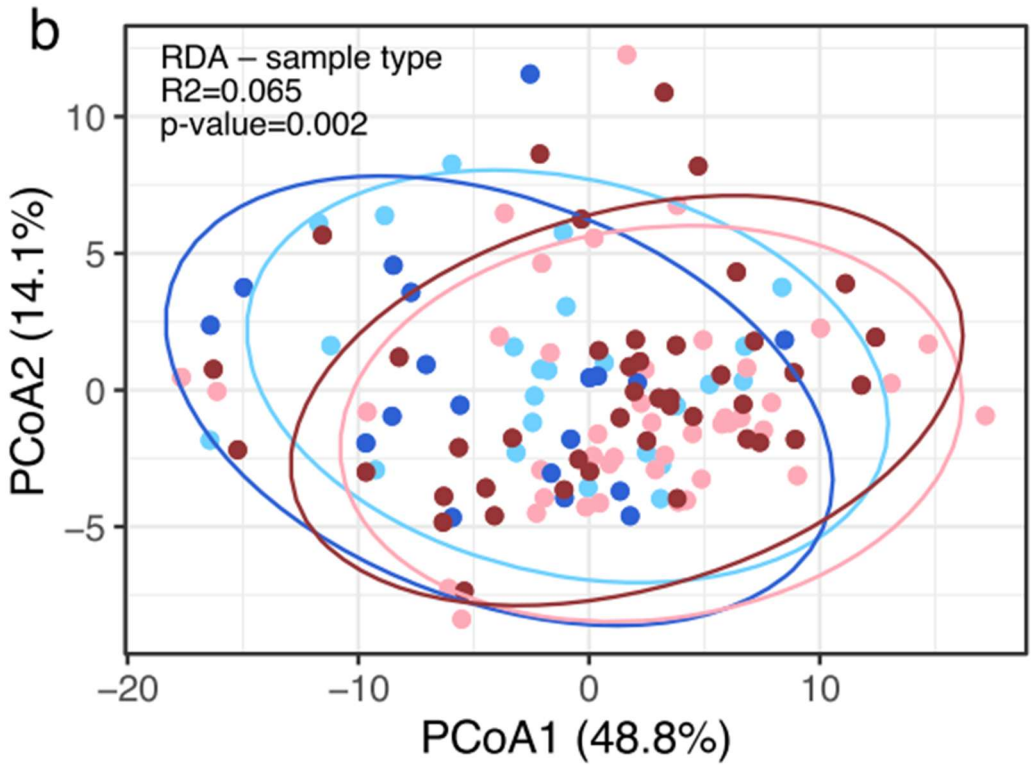




Bray-Curtis distance

- Range [0 – 1]
- No phylogenetic information

$$BC_d = \frac{\sum \left| x_i - x_j \right|}{\sum \left(x_i + x_j \right)}$$



Usually presented as a principal component plot

PROBLEM 2 – CLINICAL CASE STUDY

When routine tests are not enough

A diagnostic reasoning exercise: from blind history to mNGS-supported action

- Start with the patient, not the output.
- Build a differential before seeing mNGS results.
- Treat mNGS as evidence - not an automatic diagnosis.

Today's discipline

run first → inspect output → interpret with clinical context

Clinical mNGS should support clinical reasoning, not replace it.

Final question: Is the signal strong enough to change management today?

PROBLEM 2 – STUDENT ACTIVITY

Room setup

One server = one diagnostic team

4 groups
shared servers

10 learners
per server

2 operators
metaflow + logs

8 discussants
clinical reasoning

Operational goal: one reproducible output and one shared clinical interpretation.

Every group submits one ranked differential and one interpretation.

Rules of the Game

Start from the patient — not from the answer

- Open the clinical case instructions on the course website.
- Read the admission context, symptoms, laboratory results, imaging clues, and sample instructions.
- **Do not jump to the metagenomic answer before your team has made a clinical prediction.**
- Write down uncertainty explicitly: what fits, what conflicts, and what is missing?
- Treat the metaflow result as diagnostic evidence - not as an automatic diagnosis.

Deliverable before reveal

- **Top 5 suspected pathogens.**
- One-line rationale for each.
- Do not jump to the metagenomic answer before your team has made a clinical prediction.
- Why you cannot narrow the list
- What test/action you would choose next.

No “black box” answers: justify every conclusion.

PROBLEM 2 – STUDENT ACTIVITY

Four samples used in the manual guide

Use only paired-end FASTQ files: 4 run accessions × 2 mates = 8 files

PRJNA234452 → **paired-end only** | **4 RUNS** → **8 FASTQ FILES**

Expected pattern: SRRxxxxxxx_1.fastq.gz + SRRxxxxxxx_2.fastq.gz for each run.

SRR1145846 — PRIMARY CNS SPECIMEN

Patient: untreated CSF

- Closest compartment for meningoencephalitis
- Untreated aliquot preserves broad microbial DNA signal
- Most clinically relevant sample to evaluate first

SRR1145844 — SAME CSF, DIFFERENT PREP

Patient: DNase-treated CSF

- CSF split after collection for an alternate library condition
- DNase treatment reduces free DNA and may enrich protected viral nucleic acids
- Useful comparator, but bacterial species calls may be harder

SRR1145845 — SYSTEMIC COMPARATOR

Patient: serum

- Asks whether a candidate signal is also present in blood
- Helps distinguish CNS-localized vs disseminated infection
- Weak/negative serum does not exclude a CNS pathogen

SRR1145847 — BACKGROUND / QC

Control: serum

- Serum control from an unrelated returning traveler in the dataset
- Flags reagent, index, environmental, or database background
- Protects against over-calling low-level contaminants

Student check: exclude the single-end files; keep exactly SRR1145844/45/46/47 with both _1 and _2 FASTQ mates.

PROBLEM 2 – STUDENT ACTIVITY

Phase 1: Clinical reasoning sprint

0–20 minutes | No sequencing results yet

GROUP TASK **Produce a ranked differential diagnosis before seeing the mNGS result.**

1. List top 5 agents

- Name the suspected pathogen or pathogen group
- Rank by clinical probability
- Do not overfit to one clue

2. Explain the evidence

- Which symptoms/labs support it?
- Which exposure or host factors support it?
- What argues against it?

3. State uncertainty

- Why can the list not be narrowed?
- Which tests are missing or slow?
- What would change management?

The goal is not to guess correctly. The goal is to reason transparently.

PROBLEM 2 – STUDENT ACTIVITY

Why the differential remains wide

0–20 minutes | No sequencing results yet

Syndromic overlap

Fever + CNS symptoms can map to many infectious and non-infectious causes.

Host factors

Immunosuppression can change disease course and test performance.

Time-dependent assays

PCR, culture, and serology depend on when the sample was taken.

Low-biomass sample

CSF can contain very little pathogen material and abundant host background.

Rare pathogens

A test not ordered is a diagnosis not made.

A broad differential is not failure; unexplained uncertainty is the failure.

Uncertainty is a clinical reality — the task is to make it explicit.

PROBLEM 2 – STUDENT ACTIVITY

Phase 2: Run metaflow

0–30 min | One team, two streams of work

Pipeline operators

- Follow instructions.
- metaflow.
- Process outputs.
- Ask for help.

Clinical reasoners

- Keep refining the top 5 list.
- Separate clinical probability from test availability.
- Prepare a three-minute group explanation.
- Decide what result would be actionable.
- Do not wait for the pipeline to think.

At minute 30: merge both streams into one diagnostic argument.

Parallel teamwork prevents ten learners from clustering around one terminal.

PROBLEM 2 – STUDENT ACTIVITY

When the output arrives

Read the signal before naming the pathogen

1. **Right compartment?**
   Is the organism detected in the clinically relevant sample?

2. **Enough support?**
   Read count, p-value, ANI, and abundance must be interpreted together.

3. **Clinical fit?**
   Does it explain the syndrome, timing, exposure, and lab pattern?

4. **Background controlled?**
   Is the signal present in the clinically relevant sample and absent from the control?

5. **Actionable?**
   Would this result change the working diagnosis or antimicrobial plan?

PROBLEM 2 – STUDENT ACTIVITY

30-minute group report

Compare the clinical prediction with the metagenomics evidence

1. Pre-mNGS top hypothesis
2. Detected agent + specimen
3. Fit with clinical picture
4. Background / noise signals
5. Next action while confirming

Moderator prompt

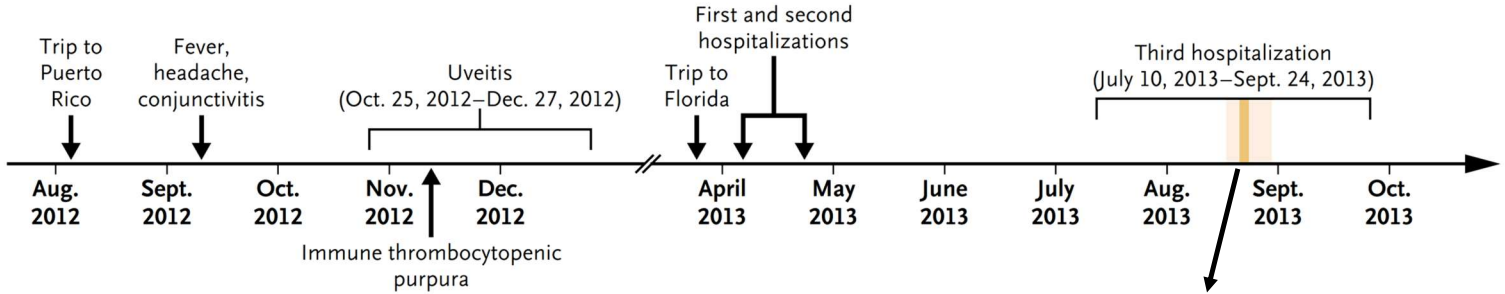
“What would make you distrust this result?”

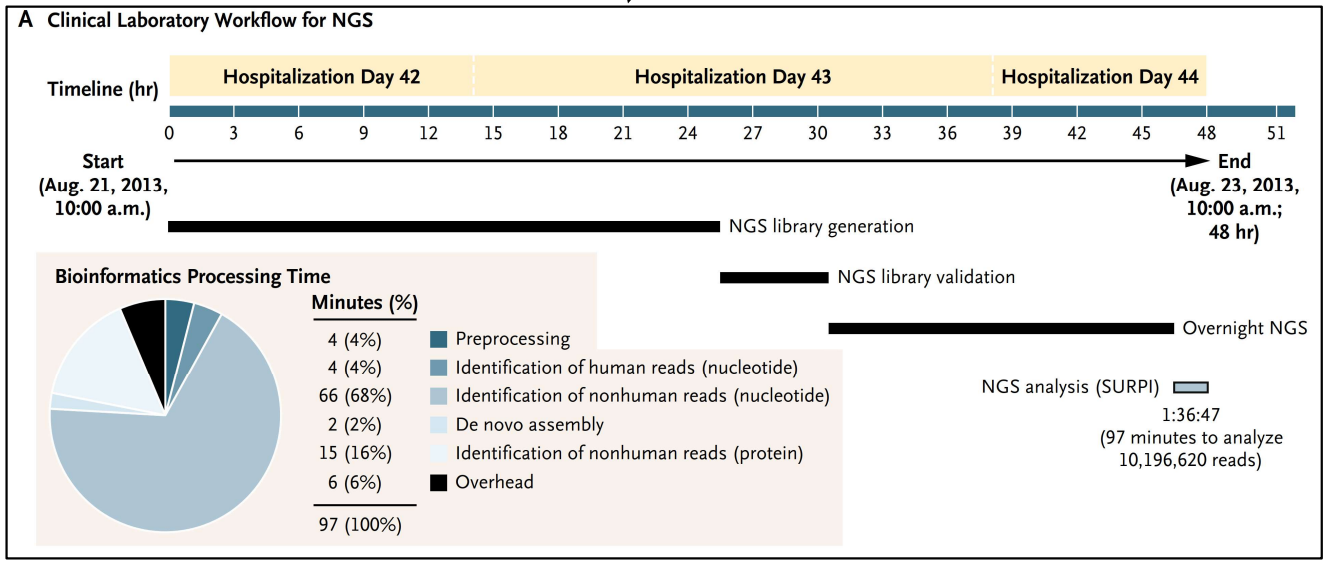
“Would you change treatment today?”

Three minutes per group: prediction → evidence → interpretation → action.

Case resolution

An actionable diagnosis of neuroleptospirosis





PROBLEM 2 – CASE REVEAL

Case resolution

An actionable diagnosis of neuroleptospirosis

The diagnostic signal

- CSF mNGS detected Leptospira reads in a severe meningoencephalitis case.
- Wilson et al. reported 475 / 3,063,784 CSF reads (0.016%) corresponding to *Leptospira*.
- The signal was later confirmed by targeted PCR/Sanger sequencing and serology as *Leptospira santarosai*.

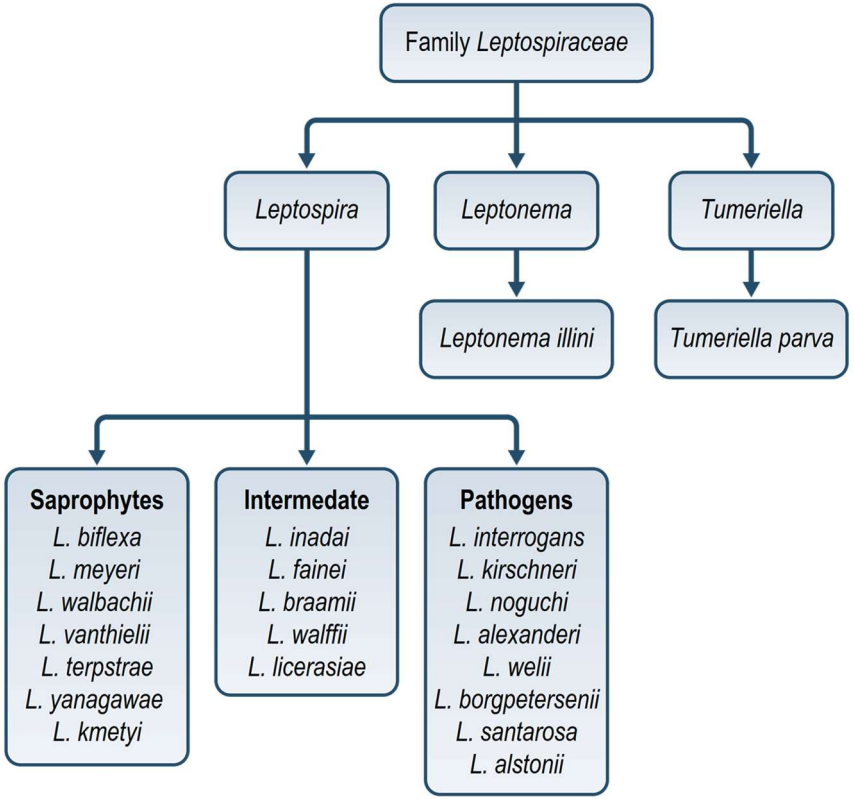
The clinical action

- Clinical leptospirosis assays were initially negative.
- The team treated before confirmatory testing was complete.
- Antibiotic coverage was narrowed to high-dose IV penicillin G.
- The patient improved and was discharged close to premorbid function.

Small fraction of reads ≠ small clinical importance.

Leptospira: Overview & Classification

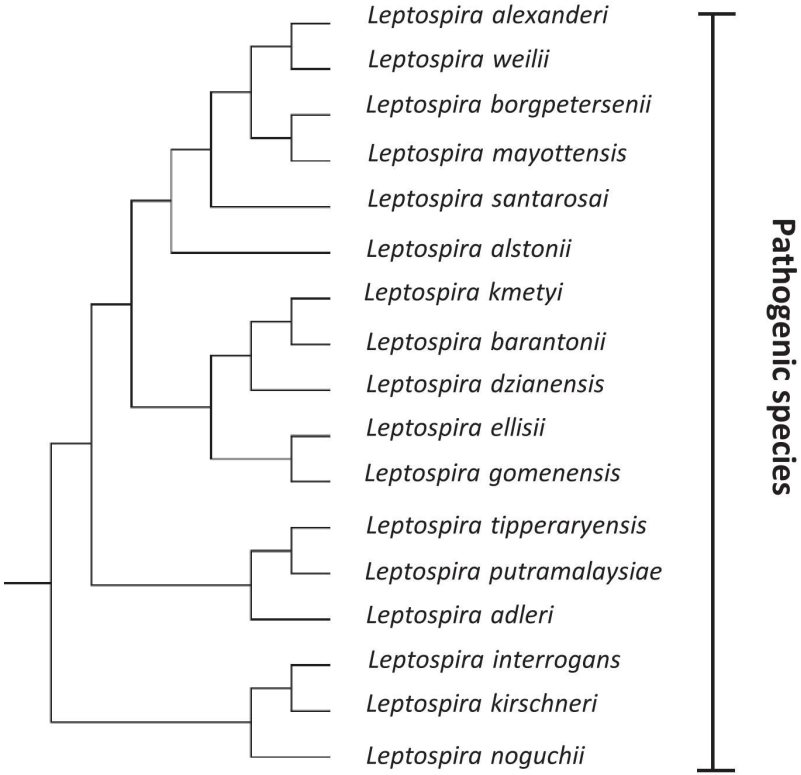
- Spiral-shaped, motile, Gram-negative bacteria
- Cause leptospirosis.
- Genomic classification: ~64-68 species
- Main groups: **pathogenic P** clade and **saprophytic S** clade
- Pathogenic species include high-virulence P1 and intermediate/uncertain P2 groups
- Bloodstream dissemination leads to leptospiremia



Classification of Leptospira. *(Manson's Tropical Diseases (24th ed.))*

Leptospira santarosai: Overview & Classification

- Pathogenic P1 species with serovars such as Shermani and Bataviae
- Reported mainly in Latin America, the Caribbean, and Taiwan
- Can cause **rare neuroleptospirosis** and **prolonged meningoencephalitis**
- **Routine MAT or PCR may be negative in difficult cases**



Differentiation of pathogenic Leptospira species by molecular phylogenetic analysis using core genomes comparison (CgMLST). *(Adapted from Harrison's Principles of Internal Medicine (22nd ed.))*

PROBLEM 2 – CASE REVEAL

Why Leptospira was a blind spot

Biologically plausible, diagnostically evasive

- **Early disease** can look like a **non-specific** febrile illness: fever, headache, myalgia, conjunctival findings.
- **Later or atypical disease** can include meningitis/encephalitis, uveitis, nephritis, and thrombocytopenia.
- Leptospires are **not well detected by routine microscopy** and **do not grow on standard culture media**.
- Serology can be **delayed**; immunosuppression and IVIG can make interpretation difficult.
- PCR can help early, but sensitivity varies with **timing, target, sample type, and Leptospira diversity**.

mNGS helps when the differential is broad and the right targeted test is unclear.

PROBLEM 2 – CASE REVEAL

Neuro-leptospirosis

Transmission

- Zoonotic spirochaete.
- Reservoir animals shed bacteria in urine.
- Entry via skin breaks or mucous membranes.
- Water, soil, flooding, agriculture, recreation.

CNS disease

- Aseptic meningitis or meningoencephalitis.
- Headache, photophobia, altered mental status.
- CSF abnormalities may include pleocytosis and high protein.
- Often not captured by routine tests.

mNGS value

- Broad, hypothesis-free search.
- Useful after negative culture/PCR/serology.
- Can support early targeted therapy.
- Still requires clinical interpretation.

Leptospirosis is not “rare” in the tropics - it is often missed because the presentation is nonspecific and testing is timing-dependent.

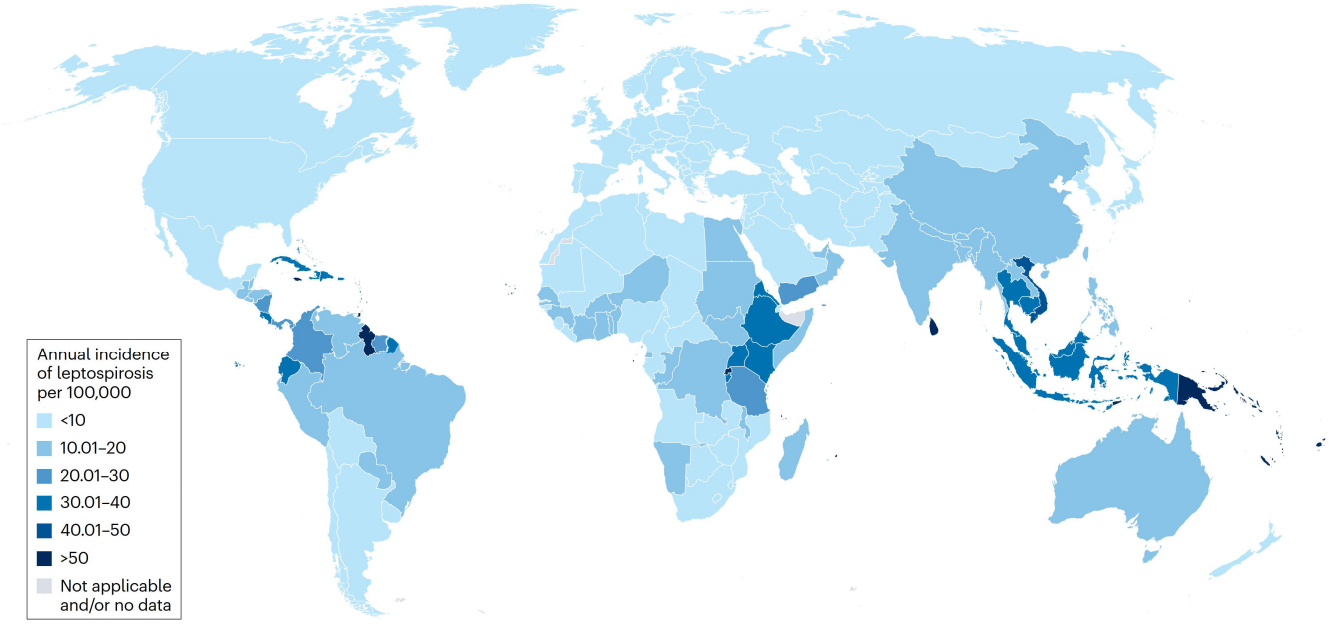
PROBLEM 2 – CASE REVEAL

Why Asia should care?

Biologically plausible, diagnostically evasive

Burden

- Leptospirosis is globally prevalent and underreported.
- High incidence in tropical and subtropical low- and middle-income settings.
- Estimated incidence is especially high in Oceania and Southeast Asia.



Estimated worldwide incidence of leptospirosis. The global estimated incidence (cases per 100,000 population) of leptospirosis per year. Darker colours indicate higher incidence. Countries with the highest rates of leptospirosis morbidity (darkest shade of blue) have a wide disease incidence range from 50 to 300 cases per 100,000 population (10.1038/s41572-025-00614-5).

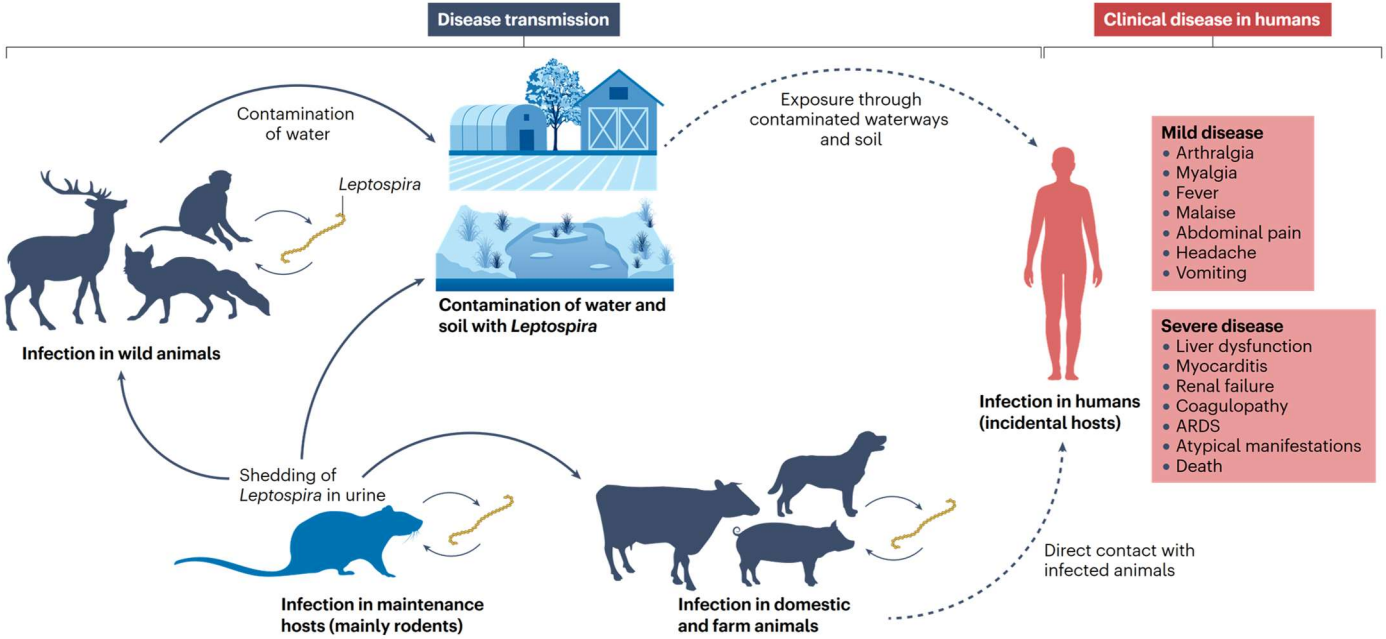
PROBLEM 2 – CASE REVEAL

Why Asia should care?

Biologically plausible, diagnostically evasive

Exposure ecology

- Rodents, livestock, and dogs can maintain transmission.
- Flooding, poor drainage, sanitation, and agricultural work increase exposure.
- Freshwater recreation and travel can also be relevant.



Transmission and clinical manifestations of leptospirosis in humans. Leptospirosis is a zoonotic infection affecting a wide range of animals, including mammals (10.1038/s41572-025-00614-5).

PROBLEM 2 – CASE REVEAL

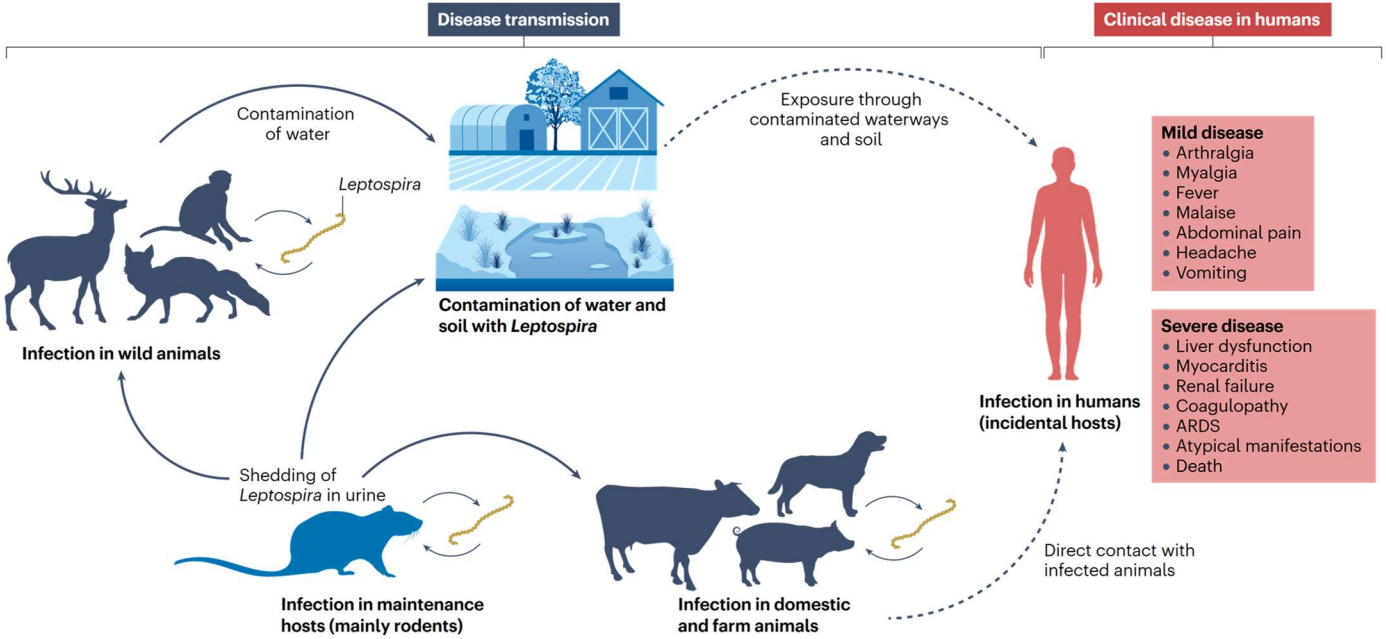
Why Asia should care?

Biologically plausible, diagnostically evasive

Asia-Oceania overlap

- Dengue, malaria, rickettsioses/scrub typhus, hantavirus, sepsis.
- The same patient may fit several syndromes.
- Species-level identification may matter epidemiologically.

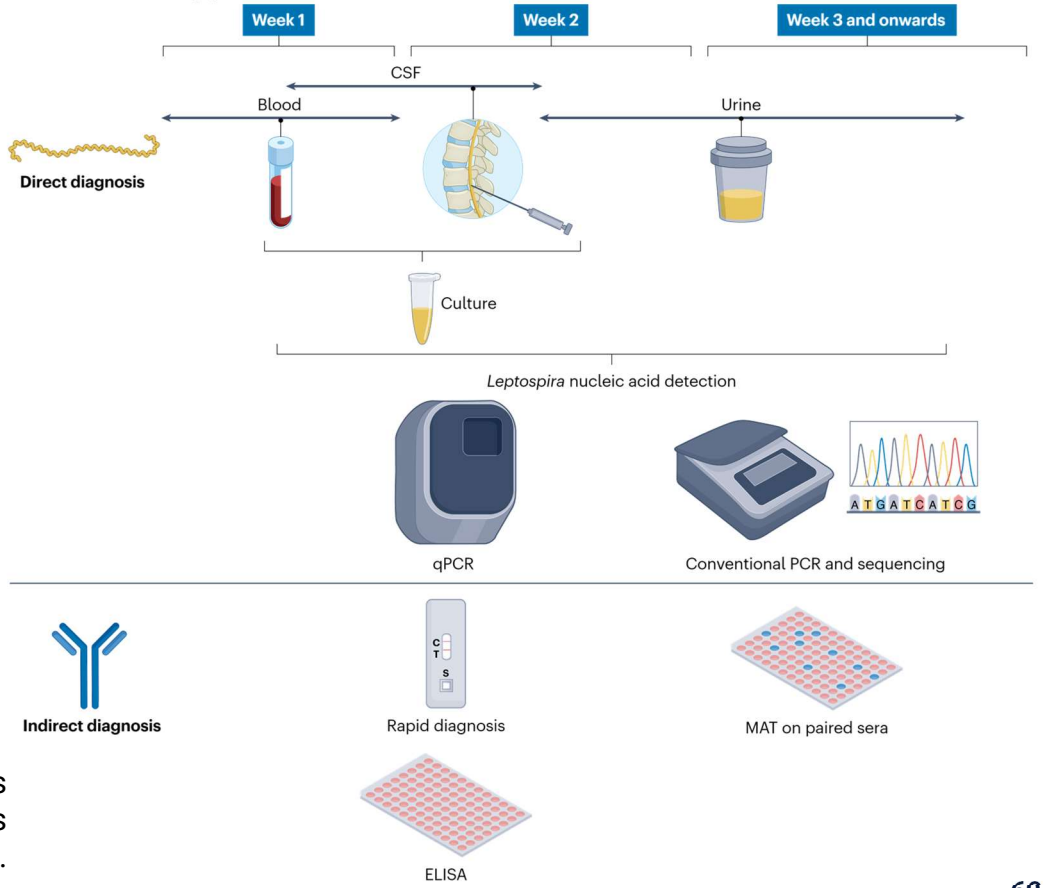
In Asia, leptospirosis should live in the differential for acute febrile illness with exposure risk - especially when tests disagree.



Transmission and clinical manifestations of leptospirosis in humans. Leptospirosis is a zoonotic infection affecting a wide range of animals, including mammals (10.1038/s41572-025-00614-5).

Conventional testing: Traditional timeline

Clinicians order tests in parallel - the bottleneck is biology and result turnaround

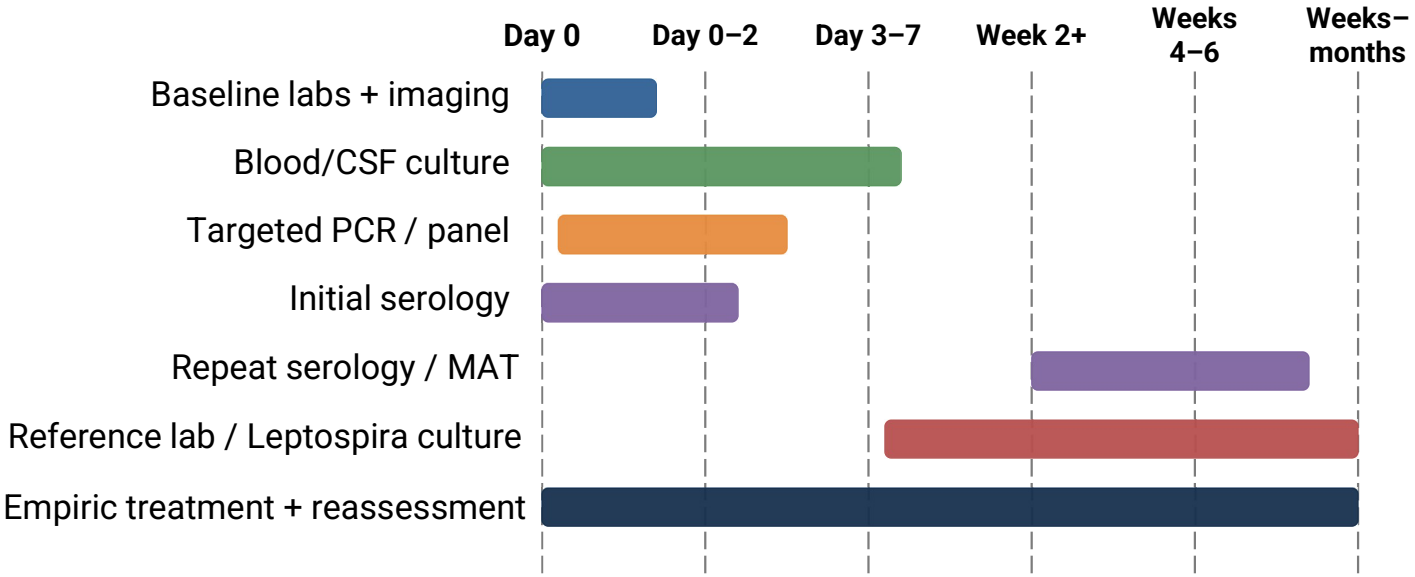


Laboratory diagnosis of leptospirosis. Leptospirosis can be detected by direct or indirect means (10.1038/s41572-025-00614-5).

PROBLEM 2 – TIME-TO-ACTION

Conventional testing: Traditional timeline

Clinicians order tests in parallel - the bottleneck is biology and result turnaround



approximate overlap map — not drawn to exact scale.

Key point

Parallel ordering saves time, but negative early tests may not rule out rare or unexpected pathogens.

Sources: Wilson et al. 2014; Rajapakse et al. 2025; Harrison. *Timings vary by local logistics*.

PROBLEM 2 – TIME-TO-ACTION

mNGS as an early parallel branch

Add broad detection while conventional assays are still running

Sample received
CSF / serum / other clinical specimen

→

Library + sequencing
Wet-lab preparation + instrument run

→

Bioinformatics
Host subtraction + microbial classification

→

Actionable report
Integrate signal with the clinical picture

Published case benchmark

- Unbiased NGS workflow: 48 h sample-to-answer.
- Bioinformatics analysis: ~97 min for >10 million reads.
- Targeted PCR/Sanger confirmation followed ~5 days later.

Decision impact

- The team started targeted penicillin G before confirmatory testing.
- Clinical recovery followed after treatment.
- mNGS converted a broad differential into an actionable hypothesis.

“what else should we test?” → “is this signal strong enough to act on today?”

References

1. Wilson, M. R., Naccache, S. N., Samayoa, E., Biagtan, M., Bashir, H., Yu, G., Salamat, S. M., Somasekar, S., Federman, S., Miller, S., Sokolic, R., Garabedian, E., Candotti, F., Buckley, R. H., Reed, K. D., Meyer, T. L., Seroogy, C. M., Galloway, R., Henderson, S. L., … Chiu, C. Y. (2014). Actionable diagnosis of neuroleptospirosis by next-generation sequencing. New England Journal of Medicine, 370(25), 2408–2417. https://doi.org/10.1056/nejmoa1401268
2. Rajapakse, S., Fernando, N., Dreyfus, A., Smith, C., & Rodrigo, C. (2025). Leptospirosis. Nature Reviews Disease Primers, 11(1). https://doi.org/10.1038/s41572-025-00614-5
3. Blaser, M. J., Cohen, J. I., Holland, S. M., Doi, Y., Falsey, A. R., Garret, W. S., Marr, K. A., Mitre, E., & Wilson, E. (2026). Mandell, Douglas, and Bennett's Principles and Practice of Infectious Diseases (10th ed., Vols. 1–2). Elsevier.
4. Farrar, J., Garcia, P. J., Hotez, P. J., Junghanss, T., Kang, G., Lalloo, D., & White, N. J. (2023). Manson's tropical diseases (24th ed.). Elsevier.
5. Longo. (2025). Harrison's Principles of Internal Medicine (22nd ed.). The McGraw-Hill Companies, Inc.

THE END.

Thanks for watching



Module 4: Shotgun metagenomics